

Reproducible code for: The rise of multidrug resistance in bacterial pathogens

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Workflow to estimate rates of AMR (anti-microbial resistance) and MDR (multi-drug resistance)

First we load necessary R packages: ape 5.7.1, phytools 1.5.1, geosphere 1.5-18, dplyr 1.1.2, tidyr 1.3.0, ggplot2 3.4.4, sf 1.0-13, rnatuarearth 0.3.4, biscale 1.0.0, cowplot 1.1.1, ggtree 3.6.2, brms 2.20.4, stringr 1.5.1, ggpubr 0.6.0, countrycode 1.5.0, raster 3.6-23, sp 2.1-1

After loading the R packages, we load two datasets (the records of resistance and a phylogeny at species level), and one function. We also introduce colour palettes for our figures. Generally, we use an increasing red palette to describe the relative multi-drug resistance. An increasing blue palette to depict the change in resistance over time.

```
library(phytools)
library(ape)
library(rnatuarearth)
library(rnatuarearthdata)
library(tidyverse)
library(countrycode)
library(geosphere)
library(rstan)
library(dplyr)
library(tidyr)
library(ggplot2)
library(sf)
library(biscale)
library(cowplot)
library(ggtree)
library(stringr)
library(raster)
library(sp)
library(ggdist)
library(ggpubr)
library(brms)
library(marginaleffects)

df_save = read.csv("DATA_AMR_jul2023_ISO3_LAT_LON 2.csv")
tr_save = read.newick("species_level_tree (2).newick")

norm_range <- function(x){
  (x-min(x, na.rm = T))/(max(x, na.rm = T)-min(x, na.rm = T))
}

palette_hot = c("#f5f5f5", "#e6b6b6", "#d87676", "#c31f1f")
palette_cold = c("#f5f5f5", "#c6c6c6", "#999999", "#6e6e6e")
palette_bi = c(
  "1-1" = "#f5f5f5", # low x, low y
  "2-1" = "#e6b6b6",
```

```

"3-1" = "#d87676",
"4-1" = "#c31f1f", # high x, low y
"1-2" = "#c6c6c6",
"2-2" = "#ba9393",
"3-2" = "#ae5f5f",
"4-2" = "#9e1919",
"1-3" = "#999999",
"2-3" = "#907272",
"3-3" = "#874949",
"4-3" = "#7a1414",
"1-4" = "#6e6e6e", # low x, high y
"2-4" = "#675252",
"3-4" = "#613535",
"4-4" = "#570e0e" # high x, high y
)

```

After loading the data, we impose a series of restrictions, removing any rows missing: coordinates, year, species name, and information on whether resistance was detected across the 13 categories. We also only keep species/countries with at least 10 observations (rows).

```

df = df_save
df = subset(df, !is.na(ISO.3)) #Remove all genomes with missing antibiotic information
df = subset(df, !is.na(centroid_lon)) #Remove all genomes with missing coordinates
df = subset(df, !is.na(centroid_lat))
print(nrow(df))

```

```
## [1] 192644
```

```

df = subset(df, !is.na(year)) #Remove all genomes with missing temporal information
print(nrow(df))

```

```
## [1] 149078
```

```

df = subset(df, !is.na(species)) #Remove all genomes with missing antibiotic information
print(nrow(df))

```

```
## [1] 149078
```

```

df = df[complete.cases(df[,c(9:21)]),] #Only keep complete cases
print(nrow(df))

```

```
## [1] 149011
```

```

geog_df = df %>%
  group_by(ISO.3) %>%
  summarise(N = n())
head(geog_df[order(-geog_df$N),]) #Countries with the most observations

```

```

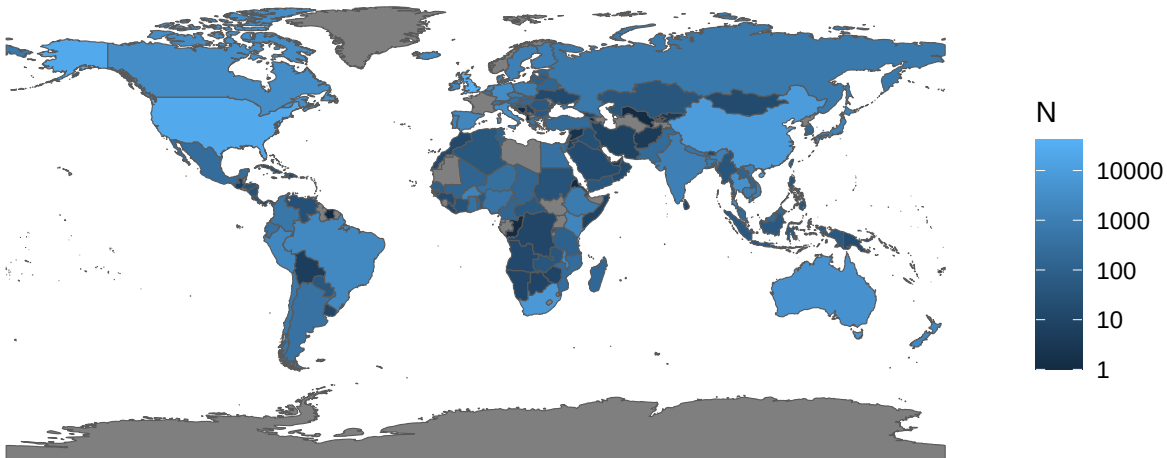
## # A tibble: 6 x 2
##   ISO.3      N
##   <chr> <int>
## 1 GBR    41584

```

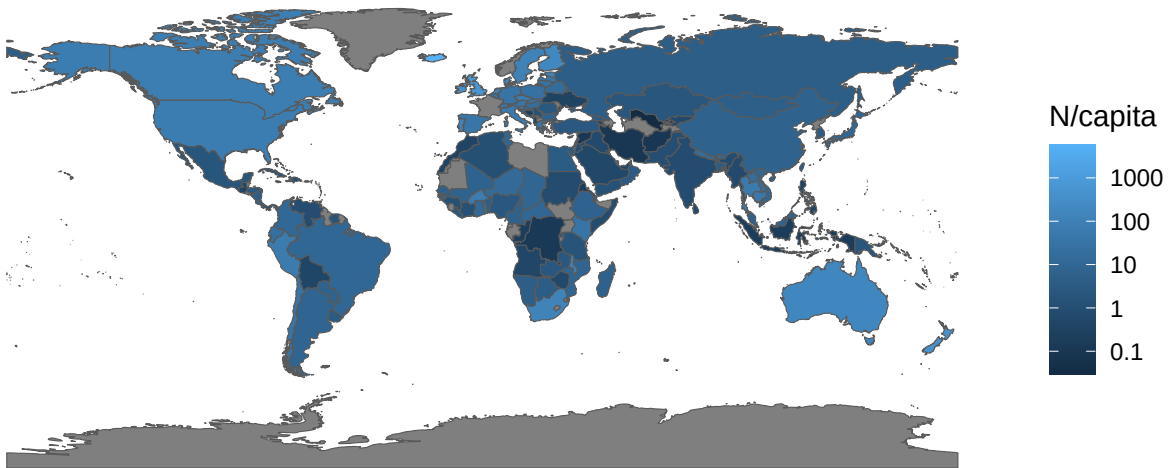
```
## 2 USA    22528
## 3 CHN    9308
## 4 ZAF    6813
## 5 AUS    4423
## 6 THA    3805
```

```
world = ne_countries(scale = "medium", returnclass = "sf")
geog_df = left_join(world, geog_df, by = c("iso_a3" = "ISO.3"))

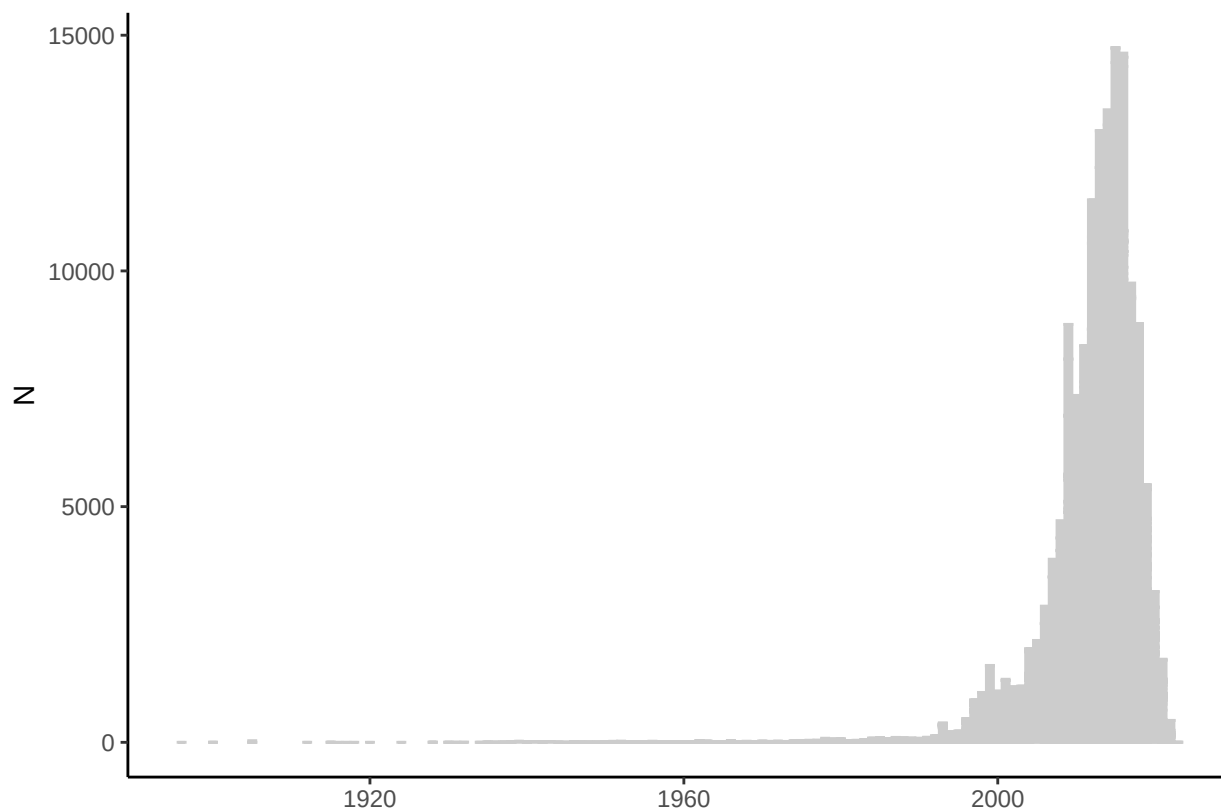
ggplot(data = geog_df) + #Plot observations over space
  geom_sf(aes(fill = N)) +
  scale_fill_gradient(name = "N", trans = "log10", breaks = c(1, 10, 100, 1000, 10000)) +
  theme_classic()
```



```
ggplot(data = geog_df) + #Samples per capita over space
  geom_sf(aes(fill = N/(pop_est/1000000))) +
  scale_fill_gradient(name = "N/capita", trans = "log10",
    breaks = c(0.1, 1, 10, 100, 1000), labels = c("0.1", "1", "10", "100", "1000")) +
  theme_classic()
```



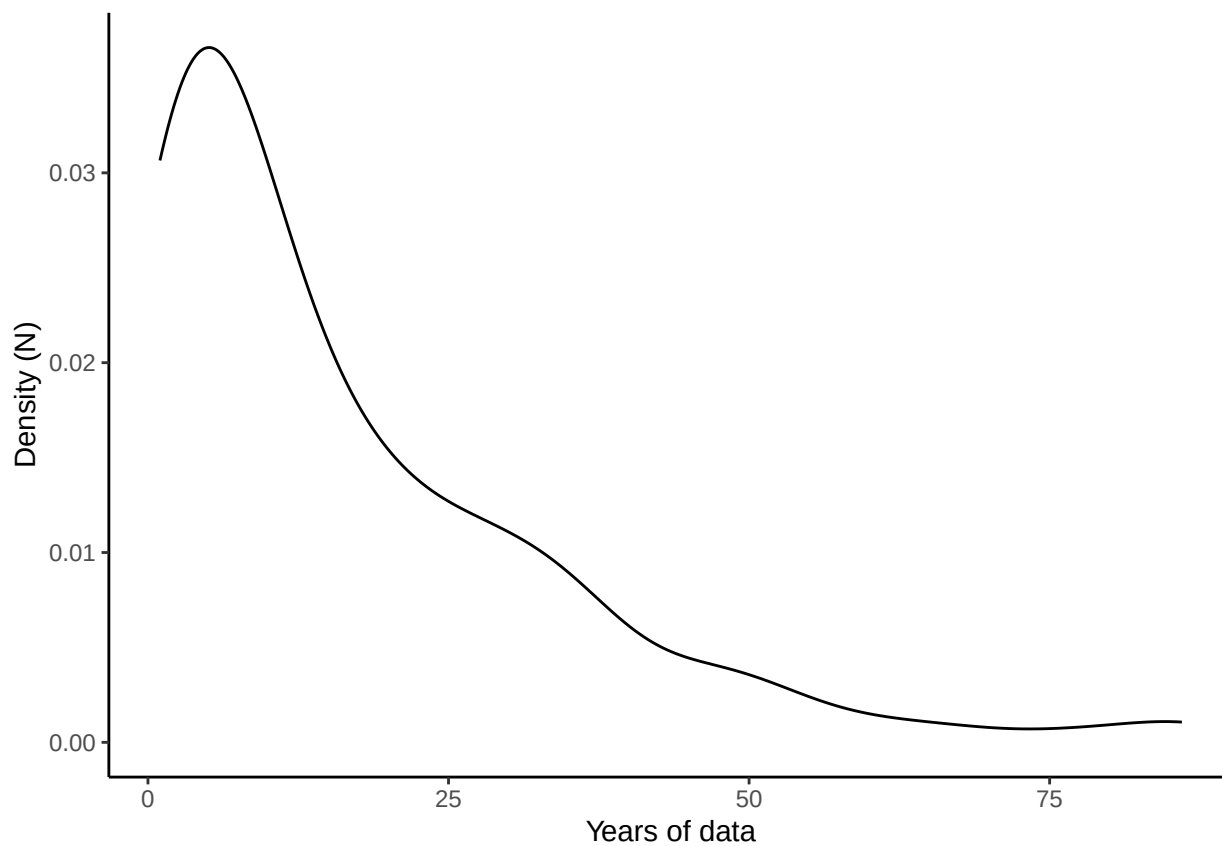
```
geog_time_df = df %>%  
  group_by(ISO.3, year) %>%  
  summarise(N = n())  
ggplot(geog_time_df) + #Number of samples over time  
  geom_col(aes(x = year, y = N), colour = "grey80", fill = "grey80") +  
  labs(x = "", y = "N") +  
  theme_classic()
```



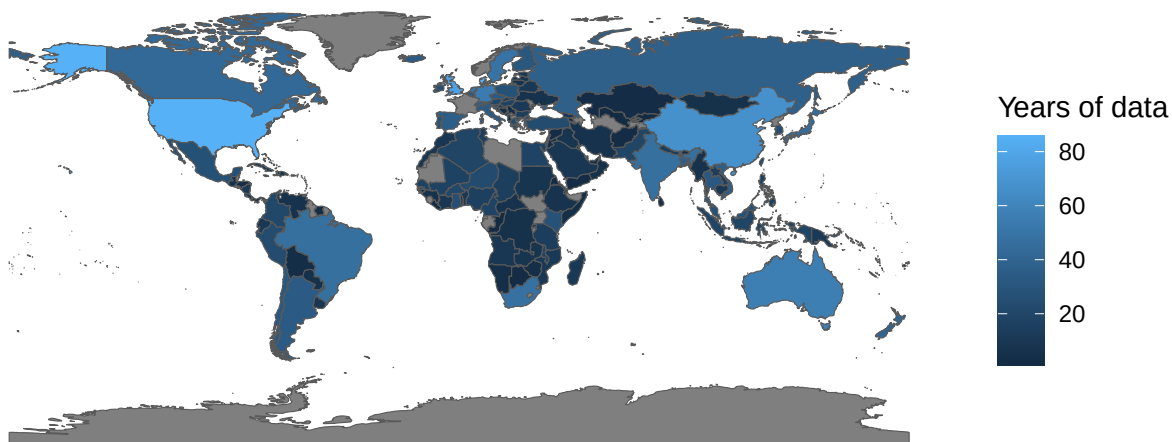
```
geog_time_counts = data.frame(sort(table(geog_time_df$ISO.3)))
length(geog_time_counts[which(geog_time_counts$Freq == 1),1]) #Countries with observations in only one year
```

```
## [1] 20
```

```
ggplot(geog_time_counts) + #Years of data per country
  geom_density(aes(x = Freq)) +
  labs(y = "Density (N)", x = "Years of data") +
  theme_classic()
```



```
geog_time_counts = left_join(world, geog_time_counts, by = c("iso_a3" = "Var1")) #Years of data over sapce
ggplot(data = geog_time_counts) +
  geom_sf(aes(fill = Freq)) +
  scale_fill_continuous(name = "Years of data") +
  theme_classic()
```



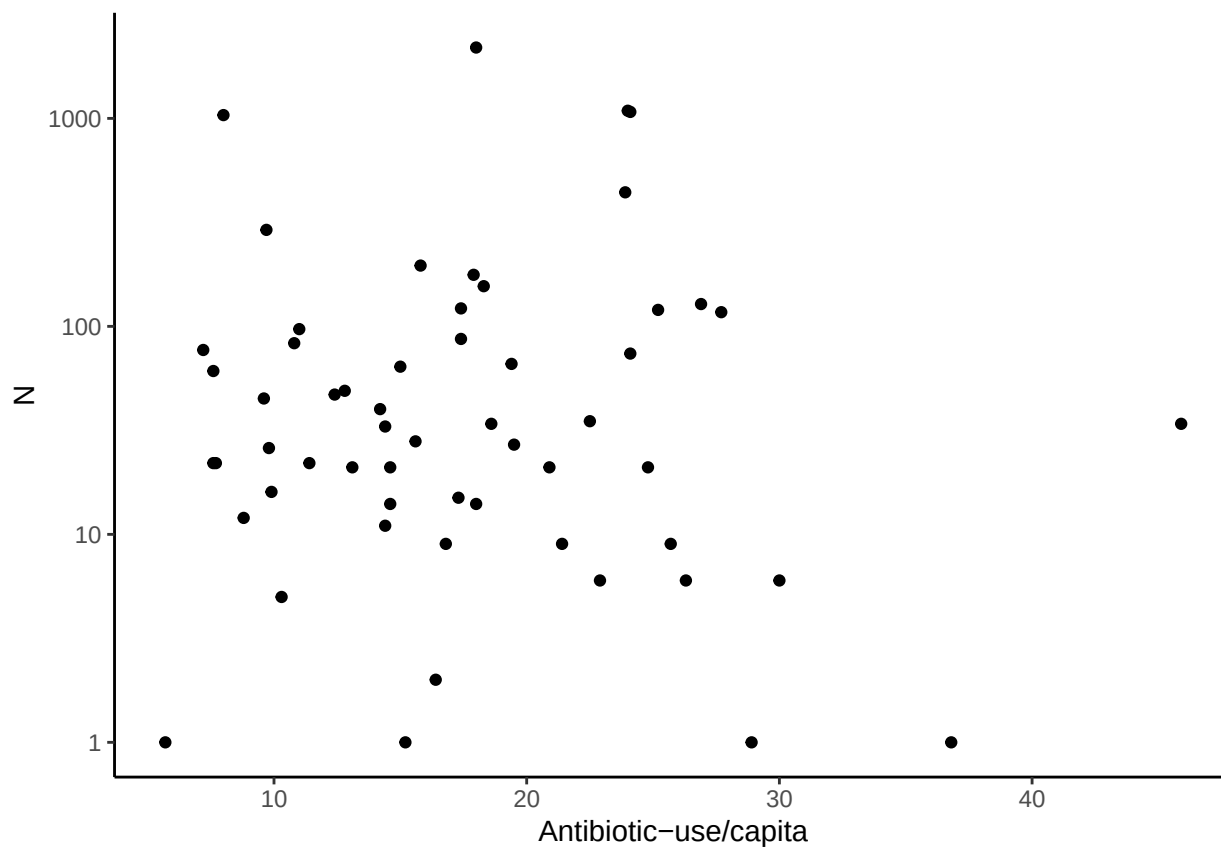
```
df = df[df$ISO.3 %in% geog_time_counts[which(geog_time_counts$Freq > 9),]$iso_a3, ] #Only keep data where there

#Load environmental data
anti = read.csv("data_anti.csv")
anti$iso3 = countrycode(anti$country, "country.name", "iso3c")
anti$country = NULL

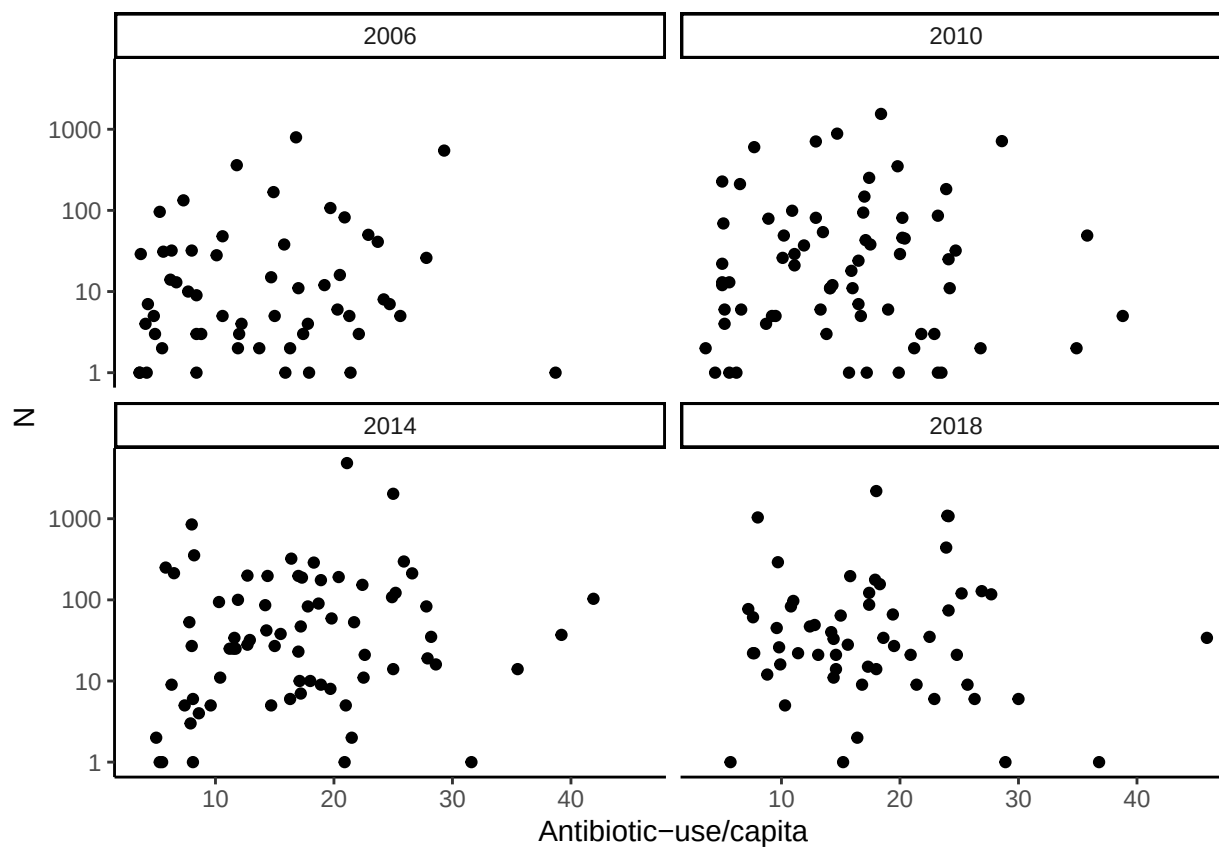
country_year_sum = df %>%
  group_by(year, ISO.3) %>%
  summarise(N = n())

country_year_sum = left_join(country_year_sum, anti, by = c("ISO.3" = "iso3", "year" = "year"))

ggplot(data = country_year_sum[which( #Plot number of samples by antibiotic use
  country_year_sum$year == 2018),]) +
  geom_point(aes(x = anti, y = N)) +
  scale_y_log10() +
  labs(x = "Antibiotic-use/capita", y = "N") +
  theme_classic()
```



```
ggplot(data = country_year_sum[which( #Plot number of samples by antibiotic use over different time periods
  country_year_sum$year == 2018 |
  country_year_sum$year == 2014 |
  country_year_sum$year == 2010 |
  country_year_sum$year == 2006),]) +
geom_point(aes(x = anti, y = N)) +
scale_y_log10() +
labs(x = "Antibiotic-use/capita", y = "N") +
theme_classic() +
facet_wrap(year~.)
```



```
species_sum = df %>%
  group_by(species) %>%
  summarise(N = n())

ggplot(species_sum) + #Number of samples per species
  geom_col(aes(x = N, y = species)) +
  scale_x_log10() +
  labs(y = "", x = "N") +
  theme_classic()
```



```

df$space_id = df$ISO.3
df$phylo_id = df$species
df$uni_id = 1:nrow(df)

# small sample
subsampldf = df
subsampldf = as.data.frame(subsampldf)
subsampldf = subsampldf %>% na.omit

GenerateMiniCov = function(df){
  #Produce a spatial covariance matrix
  spa_df = unique(df[,c("ISO.3", "centroid_lat", "centroid_lon")])
  spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000
  # probably need kernel here
  spa_mat_trim = norm_range(spa_mat_trim)
  spa_mat_trim = abs(spa_mat_trim - 1)
  spa_id = spa_df$ISO.3
  colnames(spa_mat_trim) = spa_id
  rownames(spa_mat_trim) = spa_id

  # phylo
  df = df[df$species %in% tr$tip.label, ]
  tr_trim = keep.tip(tr, unique(df$species))
  tr_trim_extend = phangorn::nnls.tree(cophenetic(tr_trim), tr_trim, rooted = TRUE)
  tr_trim_extend$edge.length = tr_trim_extend$edge.length + 0.001
  phy_mat_trim = ape::vcv.phylo(tr_trim_extend, corr = T)

  # df
  df$space_id = df$ISO.3
  df$phylo_id = df$species
  df$uni_id = 1:nrow(df)

  outlist=list(space=spa_mat_trim,phylo=phy_mat_trim,df=df)
  return(outlist)
}

standata = GenerateMiniCov(subsampldf)

```

```

#form =
# bf(mvbind(
#   Aminoglycosides,
#   Betalactams,
#   Polymyxins,
#   Fosfomicin,
#   Glycopeptides,
#   Macrolides,
#   Oxazolidinones,
#   Phenicols,
#   Quinolones,
#   Rifampicin,
#   Sulphonamides,
#   Tetracyclines,

```

```

#   Trimethoprim) ~
#   year +
#   (1 + year/ISO.3) +
#   (1 + year/species) +
#   (1/gr(space_id, cov = space)) +
#   (1/gr(phylo_id, cov = phylo))
# ) + set_rescor(FALSE)

#fit = brm(form,
#          data = standata$df,
#          data2 = list(space = standata$space, phylo = standata$phylo),
#          chains = 4, cores = 4, iter = 500, warmup = 250, thin = 1, refresh = 5,
#          family = bernoulli)

#saveRDS(fit, "non_multivar.rds")
fit = readRDS("non_multivar.rds")
summary(fit)

```

```

## Family: MV(bernoulli, bernoulli, bernoulli, bernoulli, bernoulli, bernoulli, bernoulli, bernoulli)
## Links: mu = logit
##          mu = logit
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##          mu = logit
## Formula: Aminoglycosides ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space))
##          Betalactams ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Polymyxins ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Fosfomycin ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Glycopeptides ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Macrolides ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Oxazolidinones ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Phenicolns ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Quinolones ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Rifampicin ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Sulphonamides ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Tetracyclines ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
##          Trimethoprim ~ year + (1 + year | ISO.3) + (1 + year | species) + (1 | gr(space_id, cov = space)) + (1 | gr(phylo_id, cov = phylo))
## Data: subsampled (Number of observations: 121163)
## Draws: 4 chains, each with iter = 500; warmup = 250; thin = 1;
##          total post-warmup draws = 1000
##
## Multilevel Hyperparameters:
## ~ISO.3 (Number of levels: 77)
##
##          Estimate Est.Error 1-95% CI
## sd(Aminoglycosides_Intercept)      1.80      0.21      1.43
## sd(Aminoglycosides_year)           0.16      0.02      0.13
## sd(Betalactams_Intercept)          1.35      0.18      1.03

```

## sd(Betalactams_year)	0.10	0.02	0.08
## sd(Polymyxins_Intercept)	1.96	0.47	1.14
## sd(Polymyxins_year)	0.12	0.05	0.04
## sd(Fosfomycin_Intercept)	1.54	0.34	0.95
## sd(Fosfomycin_year)	0.09	0.04	0.02
## sd(Glycopeptides_Intercept)	2.16	0.91	0.69
## sd(Glycopeptides_year)	0.13	0.10	0.01
## sd(Macrolides_Intercept)	1.52	0.20	1.19
## sd(Macrolides_year)	0.13	0.02	0.10
## sd(Oxazolidinones_Intercept)	2.23	0.87	0.41
## sd(Oxazolidinones_year)	0.37	0.12	0.19
## sd(Phenicolis_Intercept)	1.88	0.21	1.52
## sd(Phenicolis_year)	0.13	0.01	0.10
## sd(Quinolones_Intercept)	1.01	0.29	0.49
## sd(Quinolones_year)	0.09	0.02	0.05
## sd(Rifampicin_Intercept)	2.52	0.42	1.81
## sd(Rifampicin_year)	0.20	0.04	0.14
## sd(Sulphonamides_Intercept)	1.90	0.27	1.41
## sd(Sulphonamides_year)	0.11	0.02	0.08
## sd(Tetracyclines_Intercept)	1.20	0.13	0.97
## sd(Tetracyclines_year)	0.09	0.01	0.07
## sd(Trimethoprim_Intercept)	1.66	0.24	1.22
## sd(Trimethoprim_year)	0.11	0.02	0.08
## cor(Aminoglycosides_Intercept,Aminoglycosides_year)	0.82	0.06	0.70
## cor(Betalactams_Intercept,Betalactams_year)	0.81	0.06	0.68
## cor(Polymyxins_Intercept,Polymyxins_year)	0.82	0.19	0.31
## cor(Fosfomycin_Intercept,Fosfomycin_year)	0.80	0.15	0.39
## cor(Glycopeptides_Intercept,Glycopeptides_year)	0.31	0.55	-0.88
## cor(Macrolides_Intercept,Macrolides_year)	0.89	0.05	0.78
## cor(Oxazolidinones_Intercept,Oxazolidinones_year)	0.56	0.28	-0.10
## cor(Phenicolis_Intercept,Phenicolis_year)	0.92	0.03	0.85
## cor(Quinolones_Intercept,Quinolones_year)	0.62	0.23	0.04
## cor(Rifampicin_Intercept,Rifampicin_year)	0.82	0.10	0.56
## cor(Sulphonamides_Intercept,Sulphonamides_year)	0.93	0.04	0.83
## cor(Tetracyclines_Intercept,Tetracyclines_year)	0.78	0.06	0.64
## cor(Trimethoprim_Intercept,Trimethoprim_year)	0.90	0.06	0.76
##	u-95% CI	Rhat	Bulk_ESS
## sd(Aminoglycosides_Intercept)	2.25	1.05	94
## sd(Aminoglycosides_year)	0.20	1.06	92
## sd(Betalactams_Intercept)	1.74	1.01	158
## sd(Betalactams_year)	0.14	1.03	116
## sd(Polymyxins_Intercept)	2.96	1.02	184
## sd(Polymyxins_year)	0.22	1.01	116
## sd(Fosfomycin_Intercept)	2.27	1.04	130
## sd(Fosfomycin_year)	0.16	1.06	72
## sd(Glycopeptides_Intercept)	4.21	1.02	243
## sd(Glycopeptides_year)	0.39	1.02	180
## sd(Macrolides_Intercept)	1.90	1.01	334
## sd(Macrolides_year)	0.16	1.02	279
## sd(Oxazolidinones_Intercept)	3.91	1.06	76
## sd(Oxazolidinones_year)	0.61	1.02	123
## sd(Phenicolis_Intercept)	2.31	1.03	174
## sd(Phenicolis_year)	0.16	1.03	155
## sd(Quinolones_Intercept)	1.62	1.04	130
## sd(Quinolones_year)	0.14	1.04	121
## sd(Rifampicin_Intercept)	3.40	1.01	206

## sd(Rifampicin_year)	0.29 1.03	182
## sd(Sulphonamides_Intercept)	2.46 1.02	161
## sd(Sulphonamides_year)	0.15 1.01	169
## sd(Tetracyclines_Intercept)	1.47 1.01	111
## sd(Tetracyclines_year)	0.11 1.04	67
## sd(Trimethoprim_Intercept)	2.17 1.02	161
## sd(Trimethoprim_year)	0.15 1.03	160
## cor(Aminoglycosides_Intercept,Aminoglycosides_year)	0.92 1.06	75
## cor(Betalactams_Intercept,Betalactams_year)	0.91 1.03	150
## cor(Polymyxins_Intercept,Polymyxins_year)	0.99 1.02	230
## cor(Fosfomycin_Intercept,Fosfomycin_year)	0.99 1.01	211
## cor(Glycopeptides_Intercept,Glycopeptides_year)	0.98 1.01	339
## cor(Macrolides_Intercept,Macrolides_year)	0.95 1.04	140
## cor(Oxazolidinones_Intercept,Oxazolidinones_year)	0.93 1.05	107
## cor(Phenicolis_Intercept,Phenicolis_year)	0.97 1.01	188
## cor(Quinolones_Intercept,Quinolones_year)	0.90 1.06	64
## cor(Rifampicin_Intercept,Rifampicin_year)	0.95 1.03	165
## cor(Sulphonamides_Intercept,Sulphonamides_year)	0.99 1.04	131
## cor(Tetracyclines_Intercept,Tetracyclines_year)	0.89 1.04	96
## cor(Trimethoprim_Intercept,Trimethoprim_year)	0.98 1.07	48
##	Tail_ESS	
## sd(Aminoglycosides_Intercept)	154	
## sd(Aminoglycosides_year)	177	
## sd(Betalactams_Intercept)	316	
## sd(Betalactams_year)	261	
## sd(Polymyxins_Intercept)	301	
## sd(Polymyxins_year)	211	
## sd(Fosfomycin_Intercept)	304	
## sd(Fosfomycin_year)	129	
## sd(Glycopeptides_Intercept)	465	
## sd(Glycopeptides_year)	175	
## sd(Macrolides_Intercept)	636	
## sd(Macrolides_year)	477	
## sd(Oxazolidinones_Intercept)	36	
## sd(Oxazolidinones_year)	390	
## sd(Phenicolis_Intercept)	299	
## sd(Phenicolis_year)	231	
## sd(Quinolones_Intercept)	174	
## sd(Quinolones_year)	259	
## sd(Rifampicin_Intercept)	328	
## sd(Rifampicin_year)	365	
## sd(Sulphonamides_Intercept)	199	
## sd(Sulphonamides_year)	386	
## sd(Tetracyclines_Intercept)	265	
## sd(Tetracyclines_year)	240	
## sd(Trimethoprim_Intercept)	301	
## sd(Trimethoprim_year)	258	
## cor(Aminoglycosides_Intercept,Aminoglycosides_year)	148	
## cor(Betalactams_Intercept,Betalactams_year)	237	
## cor(Polymyxins_Intercept,Polymyxins_year)	121	
## cor(Fosfomycin_Intercept,Fosfomycin_year)	125	
## cor(Glycopeptides_Intercept,Glycopeptides_year)	387	
## cor(Macrolides_Intercept,Macrolides_year)	313	
## cor(Oxazolidinones_Intercept,Oxazolidinones_year)	132	
## cor(Phenicolis_Intercept,Phenicolis_year)	356	
## cor(Quinolones_Intercept,Quinolones_year)	179	

```

## cor(Rifampicin_Intercept,Rifampicin_year)                215
## cor(Sulphonamides_Intercept,Sulphonamides_year)          265
## cor(Tetracyclines_Intercept,Tetracyclines_year)           175
## cor(Trimethoprim_Intercept,Trimethoprim_year)             174
##
## ~phylo_id (Number of levels: 64)
##
##               Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(Aminoglycosides_Intercept)      6.19      1.44      3.43      9.30 1.06
## sd(Betalactams_Intercept)          9.88      1.97      6.65     14.11 1.03
## sd(Polymyxins_Intercept)           4.53      1.26      2.57      7.47 1.00
## sd(Fosfomycin_Intercept)          10.38      1.96      7.31     14.88 1.02
## sd(Glycopeptides_Intercept)        2.14      1.72      0.07      6.28 1.01
## sd(Macrolides_Intercept)           6.24      1.03      4.36      8.38 1.04
## sd(Oxazolidinones_Intercept)       5.10      2.54      0.89     10.78 1.04
## sd(Phenicolis_Intercept)           3.27      1.05      0.77      5.19 1.08
## sd(Quinolones_Intercept)           7.20      1.97      4.20     11.51 1.13
## sd(Rifampicin_Intercept)           4.47      1.19      2.68      7.35 1.01
## sd(Sulphonamides_Intercept)        4.57      1.00      3.12      6.86 1.01
## sd(Tetracyclines_Intercept)        5.54      1.28      3.29      8.50 1.04
## sd(Trimethoprim_Intercept)         5.44      1.03      3.82      7.71 1.05
##
##               Bulk_ESS Tail_ESS
## sd(Aminoglycosides_Intercept)       66      38
## sd(Betalactams_Intercept)          119     294
## sd(Polymyxins_Intercept)           196     555
## sd(Fosfomycin_Intercept)           169     336
## sd(Glycopeptides_Intercept)        184     349
## sd(Macrolides_Intercept)            88      88
## sd(Oxazolidinones_Intercept)        55      38
## sd(Phenicolis_Intercept)            43      29
## sd(Quinolones_Intercept)            31     136
## sd(Rifampicin_Intercept)            216     341
## sd(Sulphonamides_Intercept)         192     407
## sd(Tetracyclines_Intercept)         76     143
## sd(Trimethoprim_Intercept)          90     182
##
## ~space_id (Number of levels: 77)
##
##               Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(Aminoglycosides_Intercept)       0.97      0.64      0.10      2.51 1.21
## sd(Betalactams_Intercept)           0.22      0.19      0.01      0.68 1.01
## sd(Polymyxins_Intercept)            0.74      0.63      0.04      2.48 1.02
## sd(Fosfomycin_Intercept)            0.72      0.59      0.02      2.25 1.04
## sd(Glycopeptides_Intercept)         1.21      1.29      0.03      4.47 1.00
## sd(Macrolides_Intercept)            2.18      0.46      1.36      3.15 1.06
## sd(Oxazolidinones_Intercept)        1.74      1.38      0.14      5.32 1.07
## sd(Phenicolis_Intercept)            0.85      0.45      0.03      1.82 1.06
## sd(Quinolones_Intercept)            0.53      0.51      0.00      1.88 1.11
## sd(Rifampicin_Intercept)            1.83      0.94      0.47      4.13 1.05
## sd(Sulphonamides_Intercept)         1.90      0.49      1.07      3.00 1.03
## sd(Tetracyclines_Intercept)         0.68      0.33      0.17      1.43 1.05
## sd(Trimethoprim_Intercept)          2.66      0.75      1.31      4.11 1.04
##
##               Bulk_ESS Tail_ESS
## sd(Aminoglycosides_Intercept)       15      62
## sd(Betalactams_Intercept)           166     335
## sd(Polymyxins_Intercept)            227     304
## sd(Fosfomycin_Intercept)            115      85
## sd(Glycopeptides_Intercept)         257     307

```

```

## sd(Macrolides_Intercept)          64      242
## sd(Oxazolidinones_Intercept)      58      161
## sd(Phenicolis_Intercept)          44       25
## sd(Quinolones_Intercept)          46      115
## sd(Rifampicin_Intercept)          76      109
## sd(Sulphonamides_Intercept)       133     292
## sd(Tetracyclines_Intercept)        51       61
## sd(Trimethoprim_Intercept)        83      123
##
## ~species (Number of levels: 64)
##
##                                     Estimate Est.Error 1-95% CI
## sd(Aminoglycosides_Intercept)      1.56      0.70    0.22
## sd(Aminoglycosides_year)           0.11      0.02    0.08
## sd(Betalactams_Intercept)          1.65      0.81    0.18
## sd(Betalactams_year)               0.11      0.02    0.08
## sd(Polymyxins_Intercept)           0.96      0.59    0.04
## sd(Polymyxins_year)               0.10      0.05    0.02
## sd(Fosfomycin_Intercept)           1.23      0.83    0.06
## sd(Fosfomycin_year)               0.11      0.03    0.06
## sd(Glycopeptides_Intercept)        1.88      1.15    0.15
## sd(Glycopeptides_year)            0.38      0.26    0.03
## sd(Macrolides_Intercept)           0.89      0.52    0.17
## sd(Macrolides_year)               0.10      0.02    0.07
## sd(Oxazolidinones_Intercept)       2.24      1.40    0.14
## sd(Oxazolidinones_year)           0.40      0.26    0.10
## sd(Phenicolis_Intercept)          2.23      0.68    1.11
## sd(Phenicolis_year)               0.15      0.02    0.11
## sd(Quinolones_Intercept)           1.57      0.81    0.15
## sd(Quinolones_year)               0.13      0.03    0.08
## sd(Rifampicin_Intercept)           1.10      0.68    0.07
## sd(Rifampicin_year)               0.13      0.05    0.05
## sd(Sulphonamides_Intercept)        1.03      0.52    0.12
## sd(Sulphonamides_year)             0.09      0.02    0.06
## sd(Tetracyclines_Intercept)        1.68      0.62    0.62
## sd(Tetracyclines_year)            0.10      0.02    0.07
## sd(Trimethoprim_Intercept)         0.67      0.59    0.01
## sd(Trimethoprim_year)             0.10      0.02    0.07
## cor(Aminoglycosides_Intercept,Aminoglycosides_year) 0.05      0.34   -0.67
## cor(Betalactams_Intercept,Betalactams_year)         -0.04      0.44   -0.88
## cor(Polymyxins_Intercept,Polymyxins_year)           0.39      0.50   -0.73
## cor(Fosfomycin_Intercept,Fosfomycin_year)           0.46      0.44   -0.69
## cor(Glycopeptides_Intercept,Glycopeptides_year)    -0.27      0.50   -0.96
## cor(Macrolides_Intercept,Macrolides_year)           0.52      0.34   -0.25
## cor(Oxazolidinones_Intercept,Oxazolidinones_year)  -0.41      0.43   -0.97
## cor(Phenicolis_Intercept,Phenicolis_year)          0.50      0.21    0.05
## cor(Quinolones_Intercept,Quinolones_year)           0.17      0.43   -0.70
## cor(Rifampicin_Intercept,Rifampicin_year)           0.53      0.43   -0.61
## cor(Sulphonamides_Intercept,Sulphonamides_year)    -0.15      0.40   -0.93
## cor(Tetracyclines_Intercept,Tetracyclines_year)    -0.42      0.34   -0.93
## cor(Trimethoprim_Intercept,Trimethoprim_year)      -0.13      0.54   -0.94
##
##                                     u-95% CI Rhat Bulk_ESS
## sd(Aminoglycosides_Intercept)      3.14 1.11      39
## sd(Aminoglycosides_year)           0.16 1.01     203
## sd(Betalactams_Intercept)          3.46 1.12      30
## sd(Betalactams_year)               0.15 1.02     217
## sd(Polymyxins_Intercept)          2.22 1.02     108

```

## sd(Polymyxins_year)	0.21	1.00	127
## sd(Fosfomicin_Intercept)	3.09	1.06	61
## sd(Fosfomicin_year)	0.17	1.01	437
## sd(Glycopeptides_Intercept)	4.49	1.02	207
## sd(Glycopeptides_year)	1.00	1.05	103
## sd(Macrolides_Intercept)	2.17	1.14	22
## sd(Macrolides_year)	0.14	1.02	276
## sd(Oxazolidinones_Intercept)	5.48	1.03	89
## sd(Oxazolidinones_year)	1.03	1.02	76
## sd(Phenicolis_Intercept)	3.85	1.05	62
## sd(Phenicolis_year)	0.20	1.01	255
## sd(Quinolones_Intercept)	3.30	1.08	48
## sd(Quinolones_year)	0.20	1.01	307
## sd(Rifampicin_Intercept)	2.57	1.03	83
## sd(Rifampicin_year)	0.23	1.01	211
## sd(Sulphonamides_Intercept)	2.17	1.01	103
## sd(Sulphonamides_year)	0.14	1.02	268
## sd(Tetracyclines_Intercept)	3.04	1.14	24
## sd(Tetracyclines_year)	0.15	1.04	156
## sd(Trimethoprim_Intercept)	2.13	1.10	37
## sd(Trimethoprim_year)	0.16	1.01	359
## cor(Aminoglycosides_Intercept,Aminoglycosides_year)	0.63	1.13	28
## cor(Betalactams_Intercept,Betalactams_year)	0.83	1.28	12
## cor(Polymyxins_Intercept,Polymyxins_year)	0.98	1.05	155
## cor(Fosfomicin_Intercept,Fosfomicin_year)	0.98	1.07	55
## cor(Glycopeptides_Intercept,Glycopeptides_year)	0.80	1.01	216
## cor(Macrolides_Intercept,Macrolides_year)	0.99	1.07	62
## cor(Oxazolidinones_Intercept,Oxazolidinones_year)	0.69	1.02	122
## cor(Phenicolis_Intercept,Phenicolis_year)	0.84	1.04	99
## cor(Quinolones_Intercept,Quinolones_year)	0.92	1.20	21
## cor(Rifampicin_Intercept,Rifampicin_year)	0.99	1.04	73
## cor(Sulphonamides_Intercept,Sulphonamides_year)	0.55	1.13	29
## cor(Tetracyclines_Intercept,Tetracyclines_year)	0.31	1.14	22
## cor(Trimethoprim_Intercept,Trimethoprim_year)	0.93	1.15	23
##	Tail_ESS		
## sd(Aminoglycosides_Intercept)	65		
## sd(Aminoglycosides_year)	431		
## sd(Betalactams_Intercept)	71		
## sd(Betalactams_year)	517		
## sd(Polymyxins_Intercept)	75		
## sd(Polymyxins_year)	234		
## sd(Fosfomicin_Intercept)	40		
## sd(Fosfomicin_year)	632		
## sd(Glycopeptides_Intercept)	163		
## sd(Glycopeptides_year)	162		
## sd(Macrolides_Intercept)	80		
## sd(Macrolides_year)	474		
## sd(Oxazolidinones_Intercept)	164		
## sd(Oxazolidinones_year)	83		
## sd(Phenicolis_Intercept)	71		
## sd(Phenicolis_year)	529		
## sd(Quinolones_Intercept)	187		
## sd(Quinolones_year)	473		
## sd(Rifampicin_Intercept)	98		
## sd(Rifampicin_year)	384		
## sd(Sulphonamides_Intercept)	132		

```

## sd(Sulphonamides_year) 707
## sd(Tetracyclines_Intercept) 80
## sd(Tetracyclines_year) 233
## sd(Trimethoprim_Intercept) 72
## sd(Trimethoprim_year) 440
## cor(Aminoglycosides_Intercept,Aminoglycosides_year) 47
## cor(Betalactams_Intercept,Betalactams_year) 23
## cor(Polymyxins_Intercept,Polymyxins_year) 337
## cor(Fosfomycin_Intercept,Fosfomycin_year) 78
## cor(Glycopeptides_Intercept,Glycopeptides_year) 294
## cor(Macrolides_Intercept,Macrolides_year) 97
## cor(Oxazolidinones_Intercept,Oxazolidinones_year) 162
## cor(Phenicolis_Intercept,Phenicolis_year) 127
## cor(Quinolones_Intercept,Quinolones_year) 31
## cor(Rifampicin_Intercept,Rifampicin_year) 62
## cor(Sulphonamides_Intercept,Sulphonamides_year) 106
## cor(Tetracyclines_Intercept,Tetracyclines_year) 108
## cor(Trimethoprim_Intercept,Trimethoprim_year) 24
##
## Regression Coefficients:
##
## Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## Aminoglycosides_Intercept -1.93 1.74 -5.59 1.30 1.03 128
## Betalactams_Intercept -0.78 1.86 -4.77 2.60 1.04 91
## Polymyxins_Intercept -9.90 2.05 -14.76 -6.46 1.01 255
## Fosfomycin_Intercept -5.66 3.00 -11.93 -0.25 1.13 27
## Glycopeptides_Intercept -9.45 3.62 -17.26 -3.54 1.01 238
## Macrolides_Intercept -2.64 1.99 -6.52 1.21 1.02 171
## Oxazolidinones_Intercept -11.23 4.64 -21.31 -2.55 1.03 87
## Phenicolis_Intercept -3.97 1.26 -6.39 -1.41 1.01 209
## Quinolones_Intercept -5.90 2.41 -10.80 -1.00 1.06 74
## Rifampicin_Intercept -7.92 2.32 -12.08 -2.74 1.02 190
## Sulphonamides_Intercept -4.20 1.82 -7.84 -0.72 1.02 145
## Tetracyclines_Intercept -2.65 1.38 -5.25 0.17 1.02 141
## Trimethoprim_Intercept -3.71 2.37 -8.40 0.56 1.01 166
## Aminoglycosides_year 0.10 0.03 0.04 0.17 1.04 62
## Betalactams_year 0.01 0.03 -0.05 0.06 1.01 144
## Polymyxins_year 0.05 0.06 -0.06 0.20 1.00 245
## Fosfomycin_year 0.09 0.04 0.02 0.17 1.03 129
## Glycopeptides_year 0.78 0.48 0.12 2.00 1.05 106
## Macrolides_year 0.09 0.03 0.04 0.15 1.01 211
## Oxazolidinones_year 0.91 0.44 0.29 2.03 1.03 73
## Phenicolis_year 0.06 0.04 -0.01 0.13 1.02 112
## Quinolones_year 0.11 0.04 0.04 0.18 1.01 239
## Rifampicin_year 0.18 0.07 0.05 0.33 1.01 313
## Sulphonamides_year 0.07 0.03 0.01 0.14 1.01 168
## Tetracyclines_year 0.02 0.03 -0.02 0.08 1.02 159
## Trimethoprim_year 0.09 0.04 0.03 0.16 1.00 286
##
## Tail_ESS
## Aminoglycosides_Intercept 161
## Betalactams_Intercept 222
## Polymyxins_Intercept 355
## Fosfomycin_Intercept 42
## Glycopeptides_Intercept 311
## Macrolides_Intercept 377
## Oxazolidinones_Intercept 117
## Phenicolis_Intercept 454

```

```
## Quinolones_Intercept          103
## Rifampicin_Intercept          217
## Sulphonamides_Intercept       246
## Tetracyclines_Intercept       348
## Trimethoprim_Intercept        297
## Aminoglycosides_year          110
## Betalactams_year              293
## Polymyxins_year               293
## Fosfomicin_year               470
## Glycopeptides_year            164
## Macrolides_year               379
## Oxazolidinones_year           81
## Phenicol's_year               353
## Quinolones_year               402
## Rifampicin_year               527
## Sulphonamides_year            216
## Tetracyclines_year            268
## Trimethoprim_year              513
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Results

```
ab_list = c(
  "Aminoglycosides",
  "Betalactams",
  "Polymyxins",
  "Fosfomicin",
  "Glycopeptides",
  "Macrolides",
  "Oxazolidinones",
  "Phenicol's",
  "Quinolones",
  "Rifampicin",
  "Sulphonamides",
  "Tetracyclines",
  "Trimethoprim")

pp_list = list()
for(a in ab_list){
  pp_list[[a]] = pp_check(fit, resp = a) +
    theme_classic() +
    labs(title = a)
}

draw_n = 1000
tmp_df = data.frame(year = c(min(subsampledf$year): (0)))
draw = posterior_linpred(fit, tmp_df, ndraws = draw_n, re.form = NA, transform = T)
draw = matrix(draw, draw_n*34, 13)
meta = expand.grid(draws = c(1:draw_n), year = c(1990:2023))
draw = cbind(meta, draw)
colnames(draw)[3:ncol(draw)] = ab_list
draw_long = gather(draw, ab, prob, Aminoglycosides:Trimethoprim)
```

```

ab_probs = draw_long %>%
  group_by(ab, year) %>%
  summarise(q0.025 = quantile(prob, probs = 0.025),
            q0.05 = quantile(prob, probs = 0.05),
            q0.1 = quantile(prob, probs = 0.1),
            q0.25 = quantile(prob, probs = 0.25),
            q0.5 = quantile(prob, probs = 0.5),
            q0.75 = quantile(prob, probs = 0.75),
            q0.9 = quantile(prob, probs = 0.9),
            q0.95 = quantile(prob, probs = 0.95),
            q0.975 = quantile(prob, probs = 0.975))

```

Figure 1. Rise in anti-microbial resistance. Change in resistance over time across 13 anti-microbials. Solid lines describe the median posterior prediction, averaged across all species and countries. Error bars describe the 50%, 80%, 90% and 95% credible intervals.

#Cumulative MDR

```

breaks_fun <- function(x) {
  if(max(x) > 0.8){
    c(0.25, 0.5, 0.75)
  } else if (max(x) > 0.6){
    c(0.2, 0.4, 0.6)
  } else if (max(x) > 0.5){
    c(0.1, 0.3, 0.5)
  } else if (max(x) > 0.45){
    c(0.15, 0.3, 0.45)
  } else if (max(x) > 0.4){
    c(0.1, 0.25, 0.4)
  } else if (max(x) > 0.35){
    c(0.15, 0.25, 0.35)
  } else if (max(x) > 0.3){
    c(0.1, 0.2, 0.3)
  } else if (max(x) > 0.1){
    c(0.05, 0.1, 0.15)
  } else if (max(x) > 0.05){
    c(0.01, 0.03, 0.05)
  } else if (max(x) > 0.04){
    c(0.01, 0.025, 0.04)
  } else if (max(x) > 0.03){
    c(0.01, 0.02, 0.03)
  } else if (max(x) > 0.025){
    c(0.005, 0.015, 0.025)
  } else if (max(x) > 0.0125){
    c(0.0025, 0.0075, 0.0125)
  } else if (max(x) > 0.005){
    c(0.001, 0.003, 0.005)
  } else {
    c(0.0005, 0.001, 0.0015)
  }
}

```

```
fig1 = ggplot(ab_probs) +
```

```

geom_line(aes(x = as.numeric(year), y = q0.5), colour = "grey30") +
geom_ribbon(aes(x = as.numeric(year), ymin = q0.25, ymax = q0.75), fill = "grey30", alpha = 0.4) +
geom_ribbon(aes(x = as.numeric(year), ymin = q0.1, ymax = q0.9), fill = "grey30", alpha = 0.3) +
geom_ribbon(aes(x = as.numeric(year), ymin = q0.05, ymax = q0.95), fill = "grey30", alpha = 0.2) +
geom_ribbon(aes(x = as.numeric(year), ymin = q0.025, ymax = q0.975), fill = "grey30", alpha = 0.1) +
scale_x_continuous(breaks = c(1990, 2005, 2020), expand = c(0,0)) +
scale_y_continuous(expand = c(0,0), breaks = breaks_fun) +
facet_wrap(~factor(ab, levels=c(
  "Oxazolidinones",
  "Glycopeptides",
  "Rifampicin",
  "Quinolones",
  "Aminoglycosides",
  "Trimethoprim",
  "Macrolides",
  "Fosfomycin",
  "Sulphonamides",
  "Polymyxins",
  "Phenicol",
  "Tetracyclines",
  "Betalactams")), scales = "free", ncol = 3) +
labs(x = "Year", y = "Probability of exhibiting resistance") +
theme_classic()

```

fig1

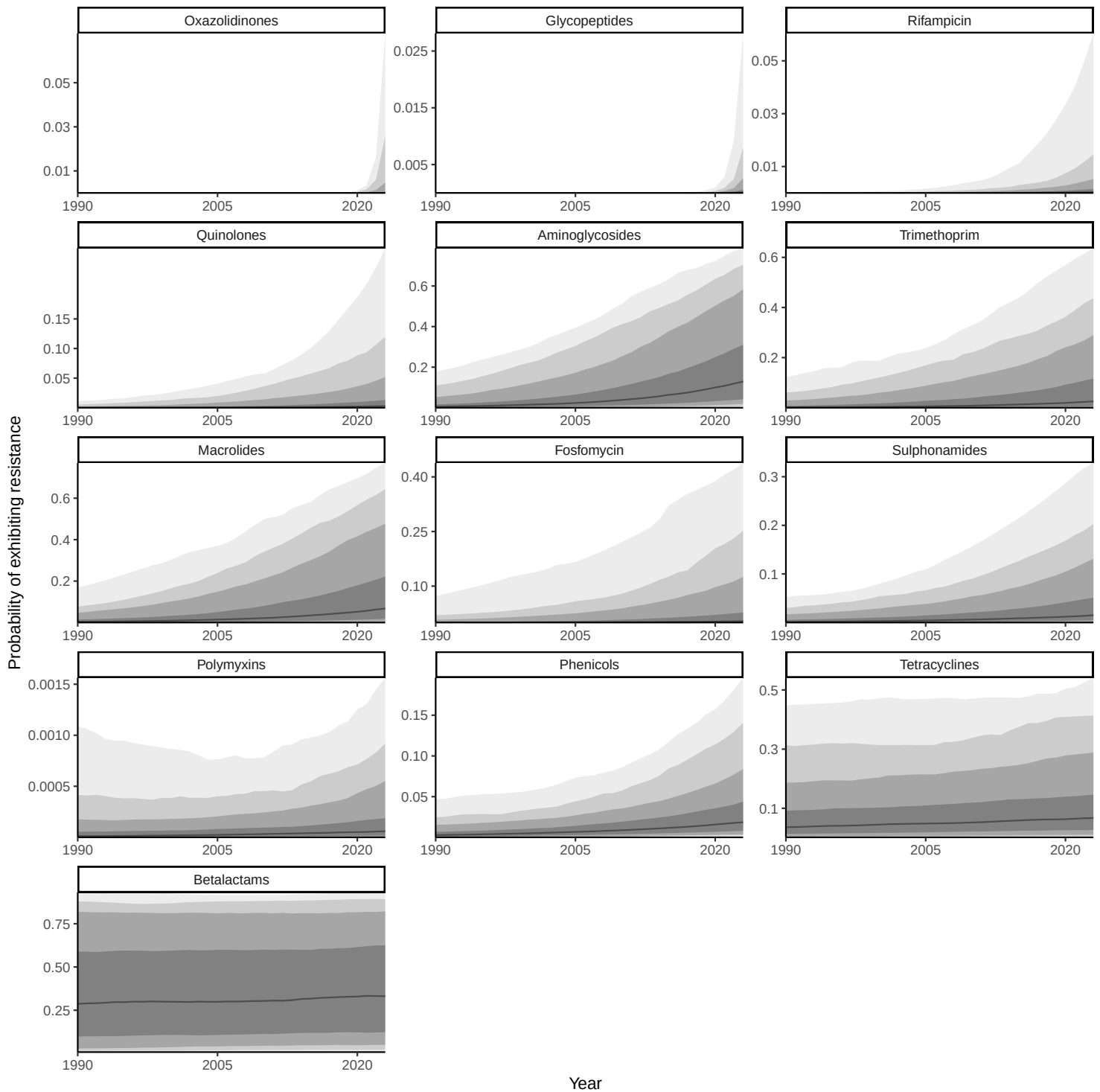


Figure 2. The arrival of multi-drug resistance. Number of resistant anti-microbials (out of a maximum of 13) over time. Proportional distribution of resistance count (N) assessed over 2000 posterior samples.

```
draw_n = 1000
tmp_df = data.frame(year = c(min(subsampledf$year): (77)))
draw = posterior_linpred(fit, tmp_df, ndraws = draw_n, re.form = NA, transform = T)
draw = matrix(draw, draw_n*111, 13)
meta = expand.grid(draws = c(1:draw_n), year = c(1990:2100))
draw = cbind(meta, draw)
```

```
colnames(draw)[3:ncol(draw)] = ab_list
draw_long = gather(draw, ab, prob, Aminoglycosides:Trimethoprim)

draw_long$bin = ifelse(draw_long$prob <= 0.5, 0, 1)
```

```
overall_mdr = draw_long %>%
  group_by(draws, year) %>%
  summarise(sum = sum(bin))

((399/326)*100)-100 #increase from 1990 to 2023 in 1 class
```

```
## [1] 22.39264
```

```
((126/7)*100)-100 #increase from 1990 to 2023 in 2 class
```

```
## [1] 1700
```

```
(17/1000)*100 #incidence of 3 classes
```

```
## [1] 1.7
```

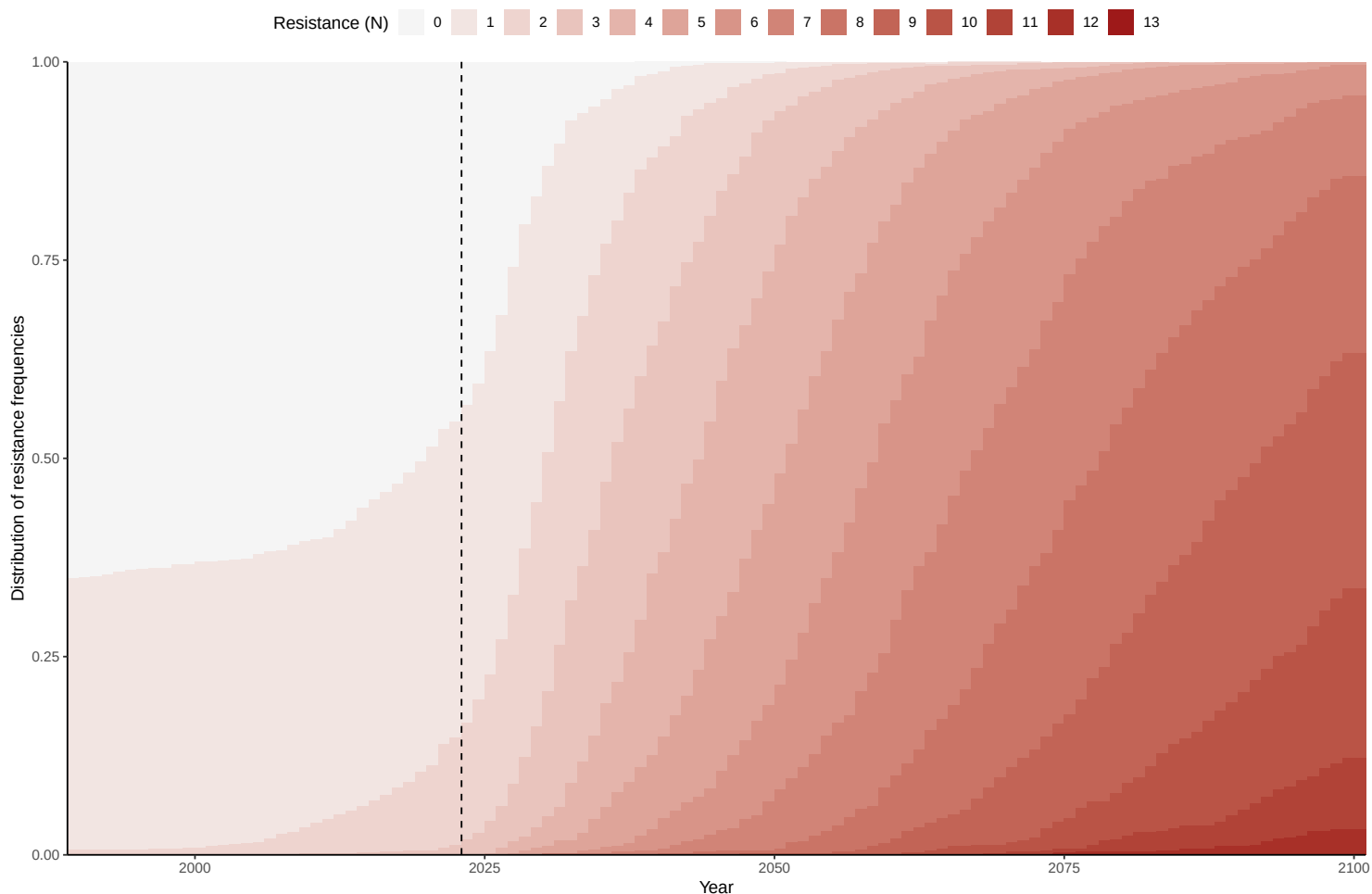
```
(7/1000)*100
```

```
## [1] 0.7
```

```
(126/1000)*100
```

```
## [1] 12.6
```

```
fig2 = ggplot(overall_mdr) +
  geom_col(aes(x = year, y = draws, fill = as.factor(sum)), width = 2, position = "fill") +
  geom_vline(aes(xintercept = 2023), linetype = "dashed") +
  scale_x_continuous(expand = c(0,0)) +
  scale_y_continuous(expand = c(0,0), breaks = c(0,0.25, 0.5, 0.75, 1)) +
  scale_fill_manual(values =
    scales::seq_gradient_pal("#f5f5f5", "#9e1919", "Lab")(seq(0,1,length.out=length(unique
  labs(x = "Year", y = "Distribution of resistance frequencies") +
  theme_classic() +
  theme(legend.position = "top") +
  guides(fill = guide_legend(nrow = 1))
fig2
```



```
tmp_df = expand.grid(year = c(-33:0), ISO.3 = unique(subsampledf$ISO.3))
tmp_df$space_id = tmp_df$ISO.3
draw = posterior_linpred(fit, tmp_df, ndraws = draw_n,
                        re.form = ~(1 + year|ISO.3) + (1|gr(space_id, cov = space)),
                        transform = T)
draw = matrix(draw, draw_n*nrow(tmp_df), 13)
meta = expand.grid(draw = c(1:draw_n), year = c(1990:2023), ISO.3 = unique(subsampledf$ISO.3))
draw = cbind(meta, draw)
colnames(draw)[4:ncol(draw)] = ab_list
draw_long = gather(draw, ab, prob, Aminoglycosides:Trimethoprim)
country_sum = draw_long %>%
  group_by(draw, year, ISO.3) %>%
  summarise(cumsum = sum(prob))
country_suma = subset(country_sum, year == 1990)
country_sumb = subset(country_sum, year == 2023)
country_sumb$diff = country_sumb$cumsum - country_suma$cumsum
country_sum2 = country_sumb %>%
  group_by(ISO.3) %>%
  summarise(mn_diff = mean(diff), sd_diff = sd(diff), mn_2023 = mean(cumsum), sd_2023 = sd(cumsum))

head(country_sum2[order(country_sum2$mn_2023, decreasing = TRUE), ] )
```

```
## # A tibble: 6 x 5
##   ISO.3 mn_diff sd_diff mn_2023 sd_2023
```

```
## <fct> <dbl> <dbl> <dbl> <dbl>
## 1 BGD 2.30 0.513 2.74 0.535
## 2 MLI 1.73 0.726 2.31 0.677
## 3 GMB 1.68 0.541 2.04 0.531
## 4 KHM 0.783 0.830 1.89 0.572
## 5 CMR 1.30 0.639 1.87 0.559
## 6 DZA 1.30 0.651 1.77 0.585
```

```
head(country_sum2[order(country_sum2$mn_2023, decreasing = FALSE), ] )
```

```
## # A tibble: 6 x 5
## ISO.3 mn_diff sd_diff mn_2023 sd_2023
## <fct> <dbl> <dbl> <dbl> <dbl>
## 1 ITA -2.42 0.573 0.348 0.283
## 2 SGP -0.536 0.383 0.475 0.281
## 3 FIN 0.00976 0.256 0.510 0.308
## 4 CAN -0.146 0.250 0.605 0.335
## 5 MDG -0.205 0.590 0.639 0.383
## 6 GBR 0.275 0.193 0.670 0.336
```

```
head(country_sum2[order(country_sum2$mn_diff, decreasing = TRUE), ] )
```

```
## # A tibble: 6 x 5
## ISO.3 mn_diff sd_diff mn_2023 sd_2023
## <fct> <dbl> <dbl> <dbl> <dbl>
## 1 BGD 2.30 0.513 2.74 0.535
## 2 MLI 1.73 0.726 2.31 0.677
## 3 GMB 1.68 0.541 2.04 0.531
## 4 BRA 1.52 0.476 1.62 0.511
## 5 PER 1.52 0.416 1.62 0.430
## 6 CMR 1.30 0.639 1.87 0.559
```

```
head(country_sum2[order(country_sum2$mn_diff, decreasing = FALSE), ] )
```

```
## # A tibble: 6 x 5
## ISO.3 mn_diff sd_diff mn_2023 sd_2023
## <fct> <dbl> <dbl> <dbl> <dbl>
## 1 ITA -2.42 0.573 0.348 0.283
## 2 CHE -0.644 0.470 0.772 0.374
## 3 NGA -0.583 0.656 0.953 0.435
## 4 SGP -0.536 0.383 0.475 0.281
## 5 KOR -0.453 0.579 0.900 0.425
## 6 THA -0.363 0.451 0.748 0.381
```

```
world_map = ne_countries(scale = "medium", returnclass = "sf")
merged_data = merge(world_map, country_sum2, by.x = "iso_a3", by.y = "ISO.3")
data = bi_class(merged_data, x = mn_2023, y = mn_diff, style = "quantile", dim = 4)
```

```
map = ggplot() +
  geom_sf(data = world_map, colour = "grey85", fill = NA) +
  geom_sf(data = data, mapping = aes(fill = bi_class), color = "white", size = 0.1, show.legend = FALSE) +
  bi_scale_fill(pal = palette_bi, dim = 4) +
  bi_theme() +
```

```

theme(plot.title = element_text(size=12))

legend = bi_legend(pal = palette_bi,
                  dim = 4,
                  xlab = "Resistance (N)",
                  ylab = "Resistance \U0394 ",
                  size = 1)

fig3a = ggdraw() +
  draw_plot(map, 0, 0, 1, 1) +
  draw_plot(legend, 0.1, 0.1, 0.35, 0.35)

#Species level MDR now and change
tmp_df = expand_grid(year = c(-33:0), species = unique(subsampledf$species))
tmp_df$phylo_id = tmp_df$species
draw = posterior_linpred(fit, tmp_df, ndraws = draw_n,
                        re.form = ~(1 + year|species) + (1|gr(phylo_id, cov = phylo)),
                        transform = T)
draw = matrix(draw, draw_n*nrow(tmp_df), 13)
meta = expand_grid(draw = c(1:draw_n), year = c(1990:2023), species = unique(subsampledf$species))
draw = cbind(meta, draw)
colnames(draw)[4:ncol(draw)] = ab_list
draw_long = gather(draw, ab, prob, Aminoglycosides:Trimethoprim)
species_sum = draw_long %>%
  group_by(draw, year, species) %>%
  summarise(cumsum = sum(prob))
species_suma = subset(species_sum, year == 1990)
species_sumb = subset(species_sum, year == 2023)
species_sumb$diff = species_sumb$cumsum - species_suma$cumsum
species_sum2 = species_sumb %>%
  group_by(species) %>%
  summarise(mn_diff = mean(diff), sd_diff = sd(diff), mn_2023 = mean(cumsum), sd_2023 = sd(cumsum))
trend_phy = setNames(species_sum2$mn_diff, species_sum2$species)
phytools::phylosig(tr, trend_phy, method = "lambda", test = T)

```

```
## [1] "some species in tree are missing from x , dropping missing taxa from the tree"
```

```
##
## Phylogenetic signal lambda : 0.999927
## logL(lambda) : -163.049
## LR(lambda=0) : 126.824
## P-value (based on LR test) : 2.03047e-29
```

```
status_phy = setNames(species_sum2$mn_2023, species_sum2$species)
phytools::phylosig(tr, status_phy, method = "lambda", test = T)

```

```
## [1] "some species in tree are missing from x , dropping missing taxa from the tree"
```

```
##
## Phylogenetic signal lambda : 0.999927
## logL(lambda) : -129.209
## LR(lambda=0) : 146.531
## P-value (based on LR test) : 9.93482e-34
```

```

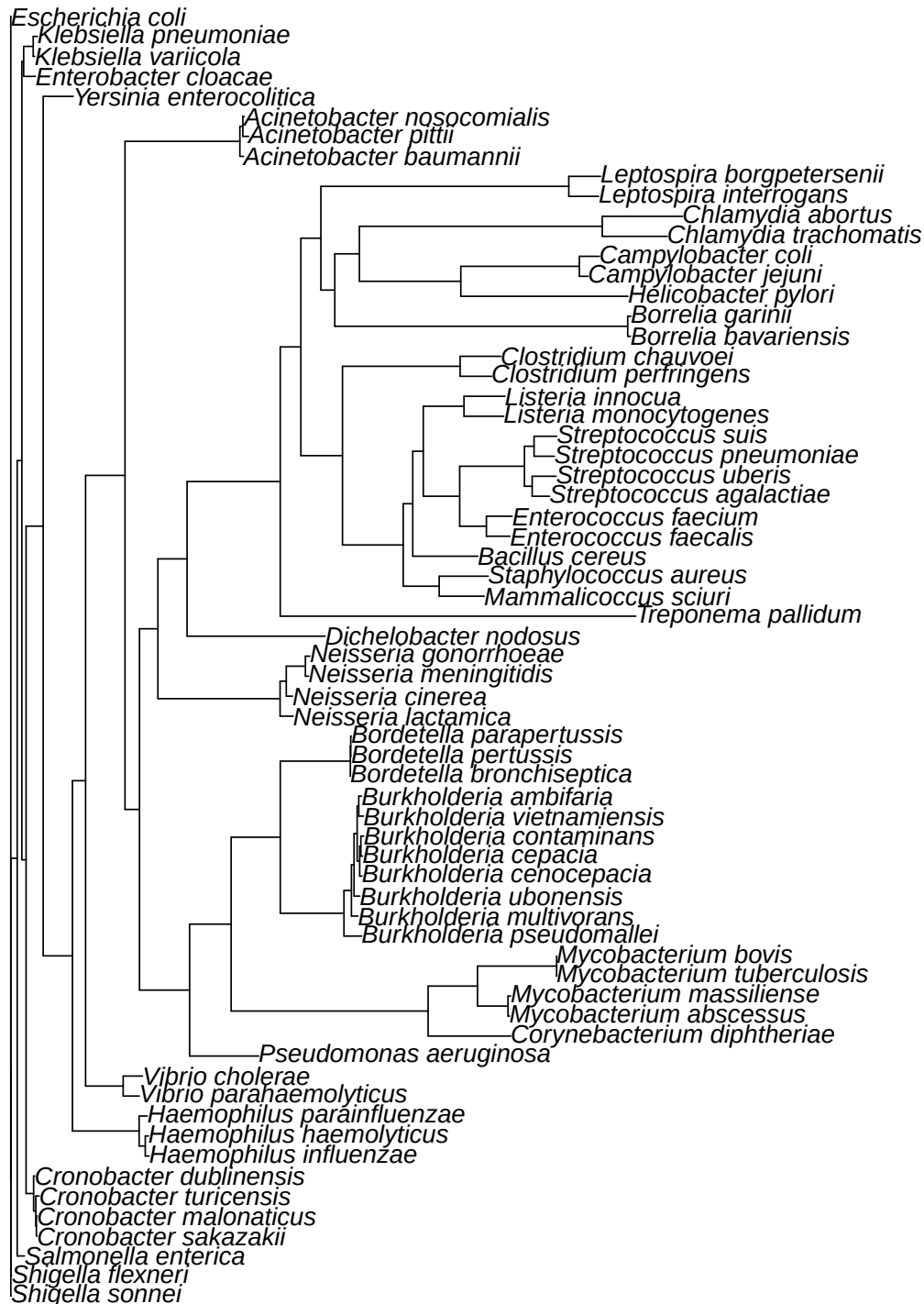
species_sum2 = bi_class(species_sum2, x = mn_2023, y = mn_diff, style = "quantile", dim = 4)[,c(1,6)]

tr = tr_save
tr_trim = keep.tip(tr, unique(subsampledf$species))

colnames(species_sum2) = c("species", "A")
species_sum2$species = gsub("_", " ", species_sum2$species)
spec_simplify = str_split_fixed(species_sum2$species, " ", 2)
species_sum2$species = paste0(substr(spec_simplify[,1], 1, 1), ". ", spec_simplify[,2])
species_sum2 = data.frame(species_sum2)
rownames(species_sum2) = species_sum2$species
species_sum2$species = NULL

tr = tr_save
tr_trim = keep.tip(tr, unique(df$species))
plot(tr_trim)

```



```

tr_trim$edge.length = log(tr_trim$edge.length) + abs(min(log(tr_trim$edge.length))) + 0.01
spec_simplify = str_split_fixed(tr_trim$tip.label, "_", 2)
tr_trim$tip.label = paste0(substr(spec_simplify[,1], 1, 1), ". ", spec_simplify[,2])
p = ggtree(tr_trim, layout = "circular") +
  geom_tiplab(align = T, size = 2)
p = gheatmap(p, species_sum2, offset = 50, width = 0.3, colnames = F) +
  scale_fill_manual(values = palette_bi, name = "") +
  theme(legend.position = "none")
fig3b = open_tree(p, angle = 20)

fig3 = ggarrange(fig3a, fig3b, heights = c(0.6, 1), ncol = 1)
fig3

```

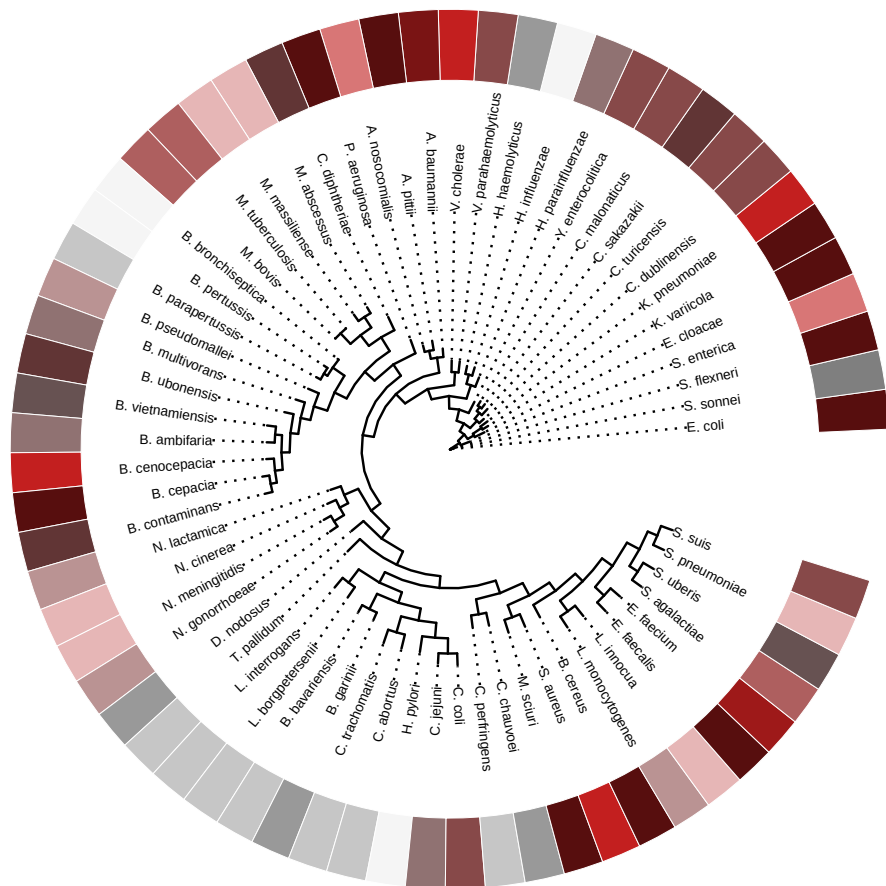
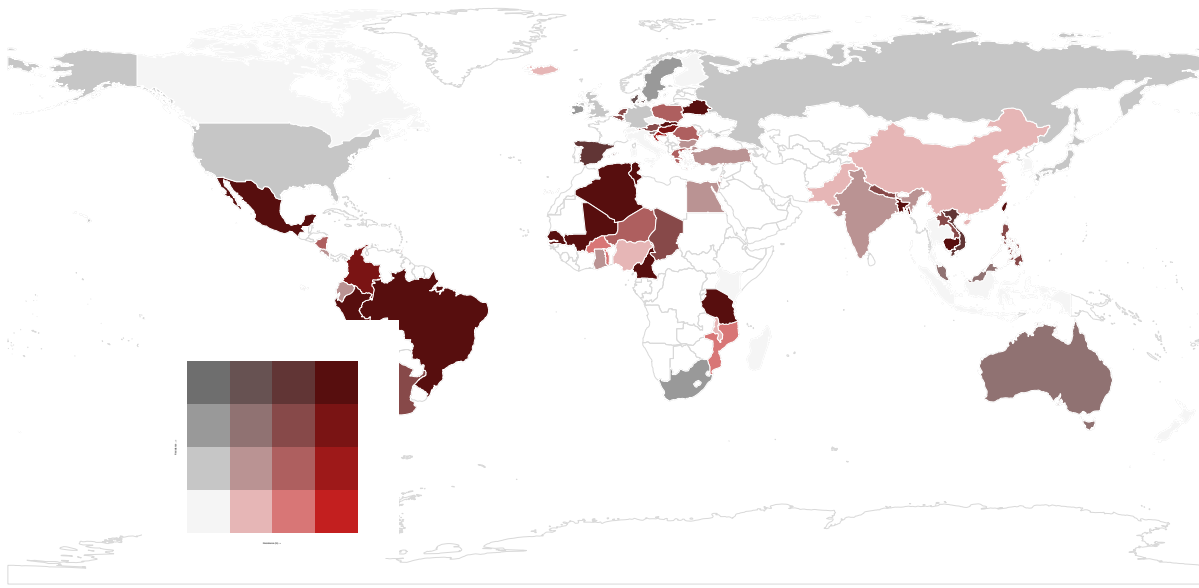


Figure 3. Distribution of resistance over space and the tree of life. A) Bi-scale map of resistance, describing the current frequency of resistance (in 2023; red) and change in resistance (between 1970 and 2023; blue). Frequency and change in resistance calculated as the cumulative probability of resistance across the 13 anti-microbials; B) Cumulative probability of resistance for each species across six decades: 1970, 1980, 1990, 2000, 2010, and 2020. To improve visulation, phylogentic branch lengths are presented on the log-scale.

Load datasets

‘data_gdp_wb.csv’. Metadata: iso3 = ISO-3 country code year = year of estimate gdp = gdp per capita at a 2015 constant

‘data_population_un.csv’. Metadata: iso3 = ISO-3 country code year = year of estimate pop = total population size as of January 1st in each year density = population density per km as July 1st in each year pop_rate = population growth rate per year, given as a percentage life_expectancy = life expectancy at birth (combined sexes) birth_mortality = mortality rate (per 1000) for children under 5 migration rate = net migration per 1000 people

data_health... Metadata: iso3 = ISO-3 country code year = year of estimate health_gdp = health expenditure as a proportion of gdp beds = hospital beds per 1,000 people physicians = physicians per 1,000 people undernourishment = percentage of population experiencing undernourishment hiv = incidence of HIV per 1,000 people malaria = incidence of malaria per 1,000 people tuberculosis = incidence of tuberculosis per 1,000 people antiretroviral = percertnage of people infected with HIV recieving treatment

data_environment... Metadata: iso3 = ISO-3 country code year = year of estimate rural = percentage of population in rural communities rural_growth = percentage growth in rural population urban = percentage of population in urban communities urban_growth = percentage growth in urban populations air_population = PM2.5 air pollution, mean annual exposure (micrograms per cubic meter) ag_land = Percentage of land occupied by agriculture

data_governance... Metadata: iso3 = ISO-3 country code year = year of estimate gini = gini coefficient of inequality corruption = control of corruption gov = government effectiveness reg = regulatory quality acc = voice and accountability

data_anti... Metadata: country = country name year = year of estimate anti = Antibiotic consumption (DDD/1,000/day) low = lower uncertainty estimate up = upper uncertainty estimate

```
#Country level MDR now and change
base = expand.grid(year = c(1990: 2023), iso3 = unique(subsampledf$ISO.3))

coords = unique(df_save[,c("ISO.3", "centroid_lat", "centroid_lon")])
coords = coords[complete.cases(coords),]
r = getData("worldclim",var="bio",res=10)[[c(1)]]
points = SpatialPoints(data.frame(x=coords$centroid_lon,y=coords$centroid_lat), proj4string = r@crs)
coords$tavg = extract(r,points)/10

population = read.csv("data_population_un.csv")
population[population == "..."] = NA

health = read.csv("data_health_wb.csv")
health[health == ".."] = NA

enviro = read.csv("data_enviro_wb.csv")
enviro[enviro == ".."] = NA

gov = read.csv("data_governance_wb.csv")
gov[gov == ".."] = NA

gdp = read.csv("data_gdp_wb.csv")
gdp[gdp == ".."] = NA

anti = read.csv("data_anti.csv")
```

```

anti$iso3 = countrycode(anti$country, "country.name", "iso3c")
anti$country = NULL

anti2 = read.csv("data_anti2.csv")[,c(1,2,5,25)]

base = left_join(base, coords, by = c("iso3" = "ISO.3"))
base = left_join(base, population)
base = left_join(base, health)
base = left_join(base, enviro)
base = left_join(base, gov)
base = left_join(base, gdp)
base = left_join(base, anti)
base = left_join(base, anti2)

miss_row = data.frame(year = base$year, iso3 = base$iso3, na = rowSums(is.na(base)))
miss_row = miss_row %>%
  group_by(year) %>%
  summarise(na = mean(na))

figs2 = ggplot(data = miss_row) +
  geom_line(aes(x = year, y = na)) +
  labs(x = "Year", y = "Mean number of missing values") +
  theme_classic() #Majority of countries possess data in the studied timeframe

base = base %>%
  group_by(iso3) %>%
  summarise(
    lat = mean(as.numeric(centroid_lat), na.rm = T),
    lon = mean(as.numeric(centroid_lon), na.rm = T),
    density = mean(as.numeric(density), na.rm = T),
    pop_rate = mean(as.numeric(pop_rate), na.rm = T),
    life_expectancy = mean(as.numeric(life_expectancy), na.rm = T),
    migration_rate = mean(as.numeric(migration_rate), na.rm = T),
    health_gdp = mean(as.numeric(health_gdp), na.rm = T),
    beds = mean(as.numeric(beds), na.rm = T),
    physicians = mean(as.numeric(physicians), na.rm = T),
    undernourishment = mean(as.numeric(undernourishment), na.rm = T),
    antiretroviral = mean(as.numeric(antiretroviral), na.rm = T),
    hiv = mean(as.numeric(hiv), na.rm = T),
    air_pollution = mean(as.numeric(air_pollution), na.rm = T),
    ag_land = mean(as.numeric(ag_land), na.rm = T),
    gini = mean(as.numeric(gini), na.rm = T),
    corruption = mean(as.numeric(corruption), na.rm = T),
    gov = mean(as.numeric(gov), na.rm = T),
    reg = mean(as.numeric(reg), na.rm = T),
    acc = mean(as.numeric(acc), na.rm = T),
    gdp = mean(as.numeric(gdp), na.rm = T),
    anti = mean(as.numeric(anti), na.rm = T),
    tavg = mean(as.numeric(tavg), na.rm = T)
  )
base = base %>% mutate_all(~ifelse(is.nan(.), NA, .))
base$gdp = log(base$gdp)
base$undernourishment = log(base$undernourishment)
base$hiv = log(base$hiv)

```

```

base$density = log(base$density)
cormat = round(cor(base[,c(4:23)], use="complete.obs"),1)
cormat = reshape2::melt(cormat)

figs3 = ggplot() +
  geom_tile(data = cormat, aes(x=Var1, y=Var2, fill=value)) +
  scale_fill_gradient2(low = scales::muted("darkblue"), high = scales::muted("darkred"), na.value = "grey20", mi
  scale_x_discrete(expand = c(0,0)) +
  scale_y_discrete(expand = c(0,0)) +
  labs(x = "", y = "") +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 90),
        legend.position = "bottom")

```

Using the posterior predictions of cumulative resistance for each country in the year 2023, we conducted a comparative analyses to explore associations between resistance and a series of predictors. These predictors covered everything from antibiotic use to governance, temperature, and quality of health care. Combined, representing 23 variables. Many of these variables are highly correlated with one another which makes inference challenging. With this in mind, we explored the link between these predictors and the response variable in a simplified linear model focussed solely on the predictors with the greatest mechanistic support: human density, proportion of GDP spent on health expenditure, governance effectiveness, antibiotic use, and the countries average temperature. We also added the following predictor interactions: health expenditure and antibiotic use, and governance effectiveness and antibiotic use. We included these in a linear model with resistance as the response variable. We included a spatial covariance term to account for covariation in neighbouring countries.

```

country_sum2 = left_join(country_sum2, base, by = c("ISO.3" = "iso3"))
country_sum2_t = gather(country_sum2, predictor, value, density:tavg)

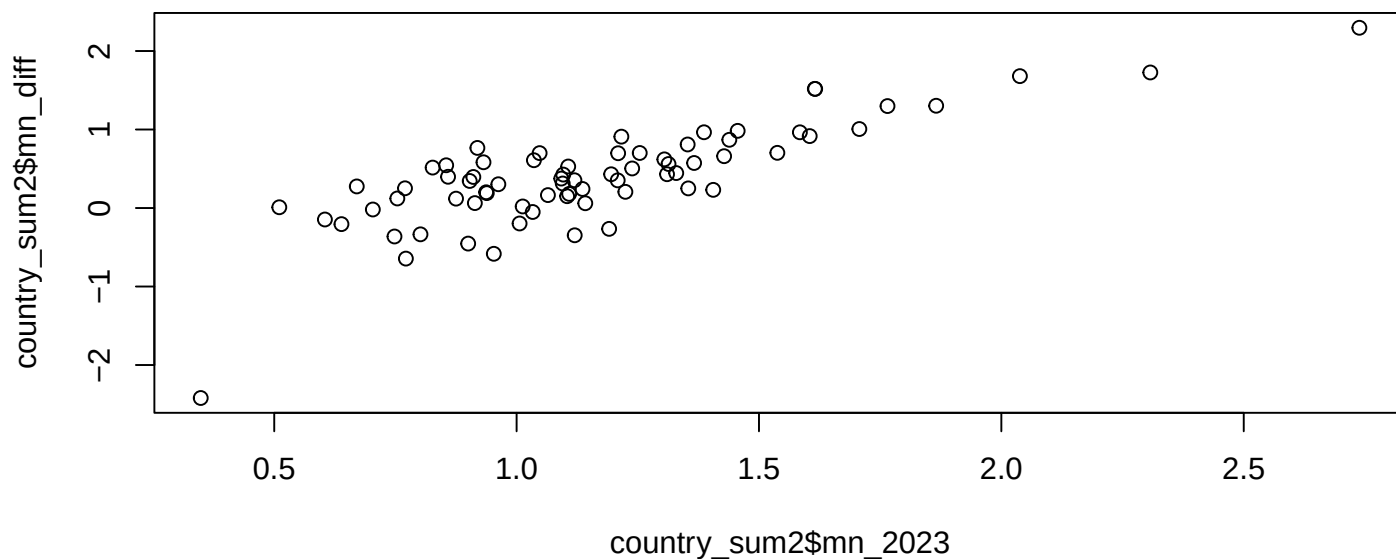
figs4 = ggplot(country_sum2_t) +
  geom_point(aes(x = value, y = mn_2023)) +
  geom_smooth(aes(x = value, y = mn_2023), method = "lm") +
  labs(x = "Value", y = "Resistance frequency in 2023") +
  facet_wrap(predictor~., scales = "free") +
  theme_classic()

country_sum2 = country_sum2[complete.cases(country_sum2[,c(
  "density",
  "health_gdp",
  "gov",
  "gdp",
  "gini",
  "anti",
  "tavg")])],]

#Produce a spatial covariance matrix
spa_df = unique(country_sum2[,c("ISO.3", "lat", "lon")])
spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000
spa_mat_trim = norm_range(spa_mat_trim)
spa_mat_trim = abs(spa_mat_trim - 1)
spa_id = spa_df$ISO.3
colnames(spa_mat_trim) = spa_id
rownames(spa_mat_trim) = spa_id
country_sum2$space_id = country_sum2$ISO.3

plot(country_sum2$mn_2023, country_sum2$mn_diff) #Highly correlated, just use absolute mdr i.e mn_2023

```



```
form_inf = bf(mn_2023~
  scale(density) +
  scale(health_gdp) +
  scale(gov) +
  scale(anti) +
  scale(tavg) +
  scale(anti)*scale(health_gdp) +
  scale(anti)*scale(gov) +
  (1|gr(space_id, cov = space)))

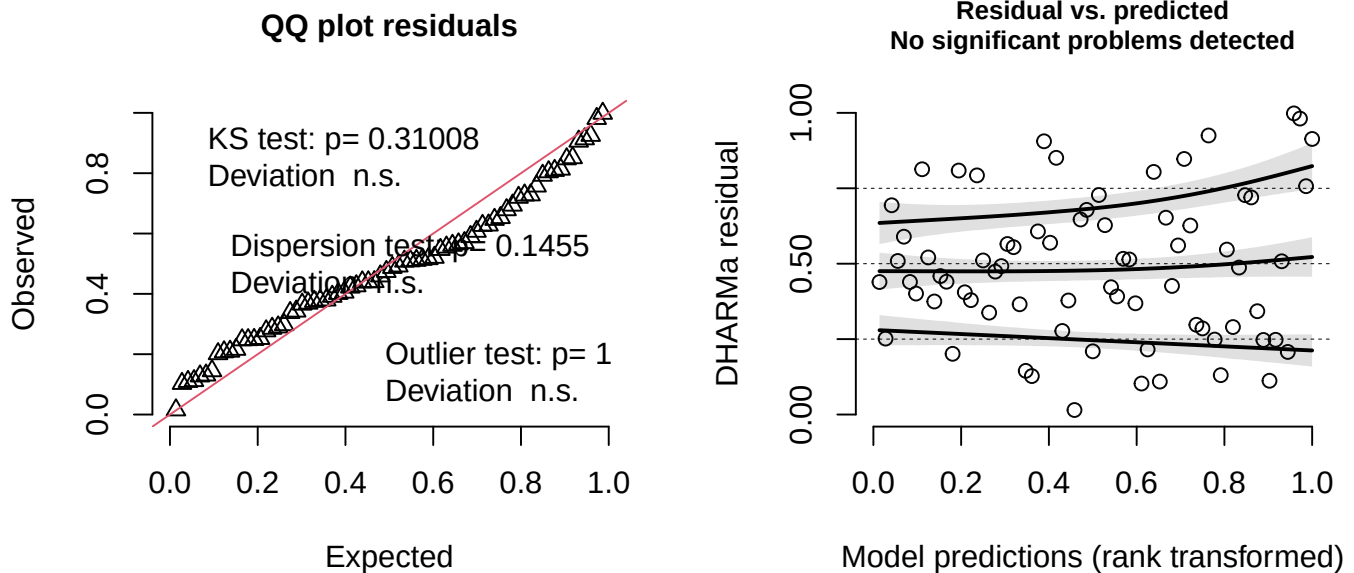
m_inf = brm(
  formula = form_inf,
  data = country_sum2,
  data2 = list(space = spa_mat_trim),
  family = gaussian,
  chains = 4, cores = 4, iter = 2000, warmup = 1000)
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG -I"/Library/Frameworks/R.f
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/StanHeaders/include
## In file included from /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/
## In file included from /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/
## /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Ma
## namespace Eigen {
## ~
## /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Ma
## namespace Eigen {
## ~
## ;
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/StanHeaders/include
```

```
## In file included from /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/
## /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/library/RcppEigen/include/Eigen/Core:96:10: fatal
## #include <complex>
##          ~~~~~~
## 3 errors generated.
## make: *** [foo.o] Error 1
```

```
figs5 = pp_check(m_inf) +
  theme_classic()
model_check = DHARMA::createDHARMA(
  simulatedResponse = (t(posterior_predict(m_inf))),
  observedResponse = country_sum2$mn_2023,
  fittedPredictedResponse = (apply(t(posterior_epred(m_inf)), 1, mean)),
  integerResponse = FALSE)
figs6 = plot(model_check)
```

DHARMA residual



```
summary(m_inf)
```

```
## Family: gaussian
## Links: mu = identity; sigma = identity
## Formula: mn_2023 ~ scale(density) + scale(health_gdp) + scale(gov) + scale(anti) + scale(tavg) + scale(anti)
## Data: country_sum2 (Number of observations: 72)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~space_id (Number of levels: 72)
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.26	0.16	0.02	0.63	1.00	964	1623

```
##
## Regression Coefficients:
```

```
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## Intercept           1.14      0.20    0.67    1.54 1.00      884
## scaledensity         0.02      0.05   -0.08    0.12 1.00     2967
## scalehealth_gdp      -0.02      0.09   -0.20    0.15 1.00     2681
## scalegov             -0.14      0.09   -0.30    0.03 1.00     2010
## scaleanti            0.04      0.06   -0.07    0.15 1.00     3371
## scaletavg            0.11      0.08   -0.04    0.26 1.00     2318
## scalehealth_gdp:scaleanti -0.03    0.07   -0.16    0.10 1.00     2398
## scalegov:scaleanti    -0.01      0.08   -0.15    0.14 1.00     2770
##               Tail_ESS
## Intercept           426
## scaledensity        2187
## scalehealth_gdp      2353
## scalegov             2259
## scaleanti            2570
## scaletavg            1781
## scalehealth_gdp:scaleanti 2821
## scalegov:scaleanti    2561
##
## Further Distributional Parameters:
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma          0.36      0.04    0.30    0.44 1.00     1968     1874
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
post_sum = as.data.frame(posterior_summary(m_inf, probs = c(0.025, 0.1, 0.25, 0.75, 0.9, 0.975))[c(2:8),])
post_sum$var = c(
  "Population density",
  "Health expenditure",
  "Governance",
  "Antibiotic-use",
  "Average temperature",
  "Health expenditure:\nAntibiotic-use",
  "Governance:\nAntibiotic-use"
)
post_sum$col = c(
  "A",
  "A",
  "B",
  "A",
  "B",
  "A",
  "A"
)

fig4a = ggplot(post_sum) +
  geom_linerange(aes(y = reorder(var, -Q2.5), xmin = Q2.5, xmax = Q97.5, colour = col), linewidth = 5, alpha = 0.6) +
  geom_linerange(aes(y = reorder(var, -Q2.5), xmin = Q10, xmax = Q90, colour = col), linewidth = 5, alpha = 0.6) +
  geom_linerange(aes(y = reorder(var, -Q2.5), xmin = Q25, xmax = Q75, colour = col), linewidth = 5, alpha = 0.8) +
  geom_vline(aes(xintercept = 0), linetype = "dashed") +
  scale_colour_manual(guide = "none", values = c("grey30", scales::muted("#9e1919"))) +
  labs(x = expression(beta), y = "") +
  theme_classic()
```

```

pred <- predictions(m_inf,
  newdata = datagrid(gov = seq(-2.5, 2.5, by = 0.05))) |>
  posterior_draws()

fig4b = ggplot(pred, aes(x = gov, y = exp(draw))) +
  stat_lineribbon() +
  scale_x_continuous(expand = c(0,0)) +
  scale_y_continuous(expand = c(0,0)) +
  scale_fill_manual(guide = "none", values = c(fill_alpha("#9e1919", 0.4), fill_alpha("#9e1919", 0.6), fill_alpha("#9e1919", 0.8))) +
  labs(x = "Government effectiveness",
    y = expression("Resistance (N)")) +
  theme_classic()

pred <- predictions(m_inf,
  newdata = datagrid(tavg = seq(0, 30, by = 0.1))) |>
  posterior_draws()

fig4c = ggplot(pred, aes(x = tavg, y = exp(draw))) +
  stat_lineribbon() +
  scale_x_continuous(expand = c(0,0)) +
  scale_y_continuous(expand = c(0,0)) +
  scale_fill_manual(guide = "none", values = c(fill_alpha("#9e1919", 0.4), fill_alpha("#9e1919", 0.6), fill_alpha("#9e1919", 0.8))) +
  labs(x = "Average temperature",
    y = expression("Resistance (N)")) +
  theme_classic()

ggarrange(fig4a, fig4b, fig4c, ncol = 3, widths = c(1,0.7,0.7))

```

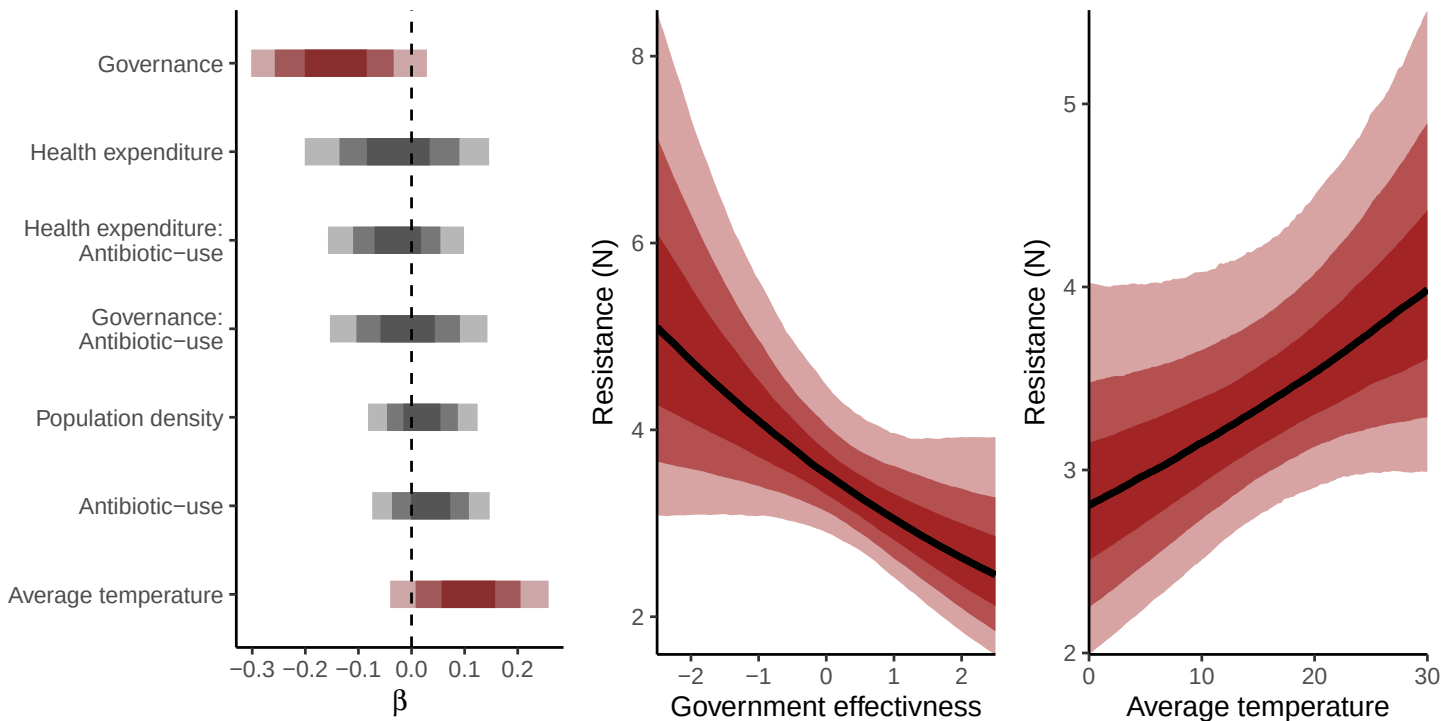


Figure 4. Human density, health expenditure, and financial inequality are linked to resistance Marginal effect posterior predictions of the association between resistance and the following predictors: human density, health expenditure, and financial inequality. Error bars describe 95% credible intervals.

Supplementary material

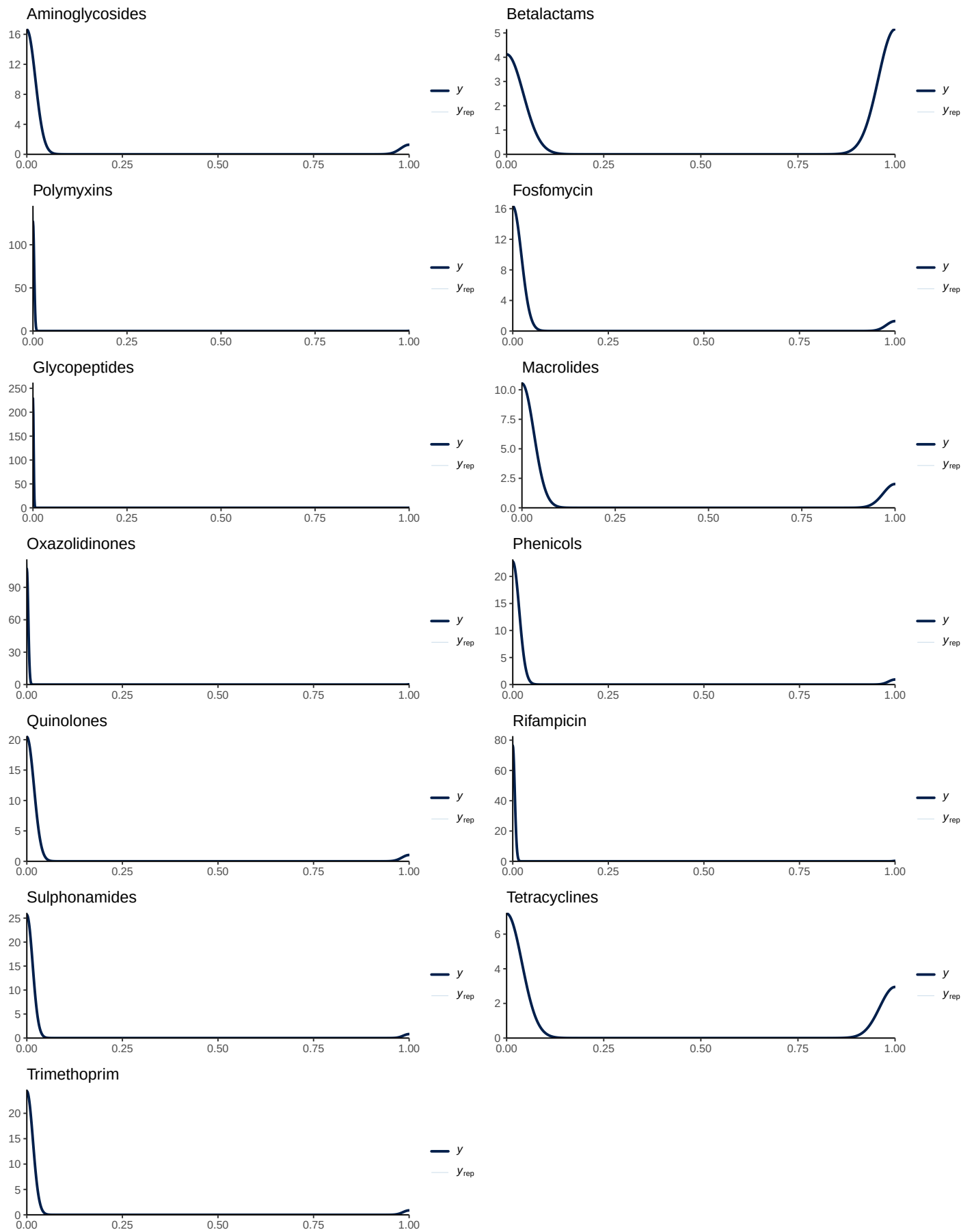


Figure S1. Posterior predictive checks of temporal trend model

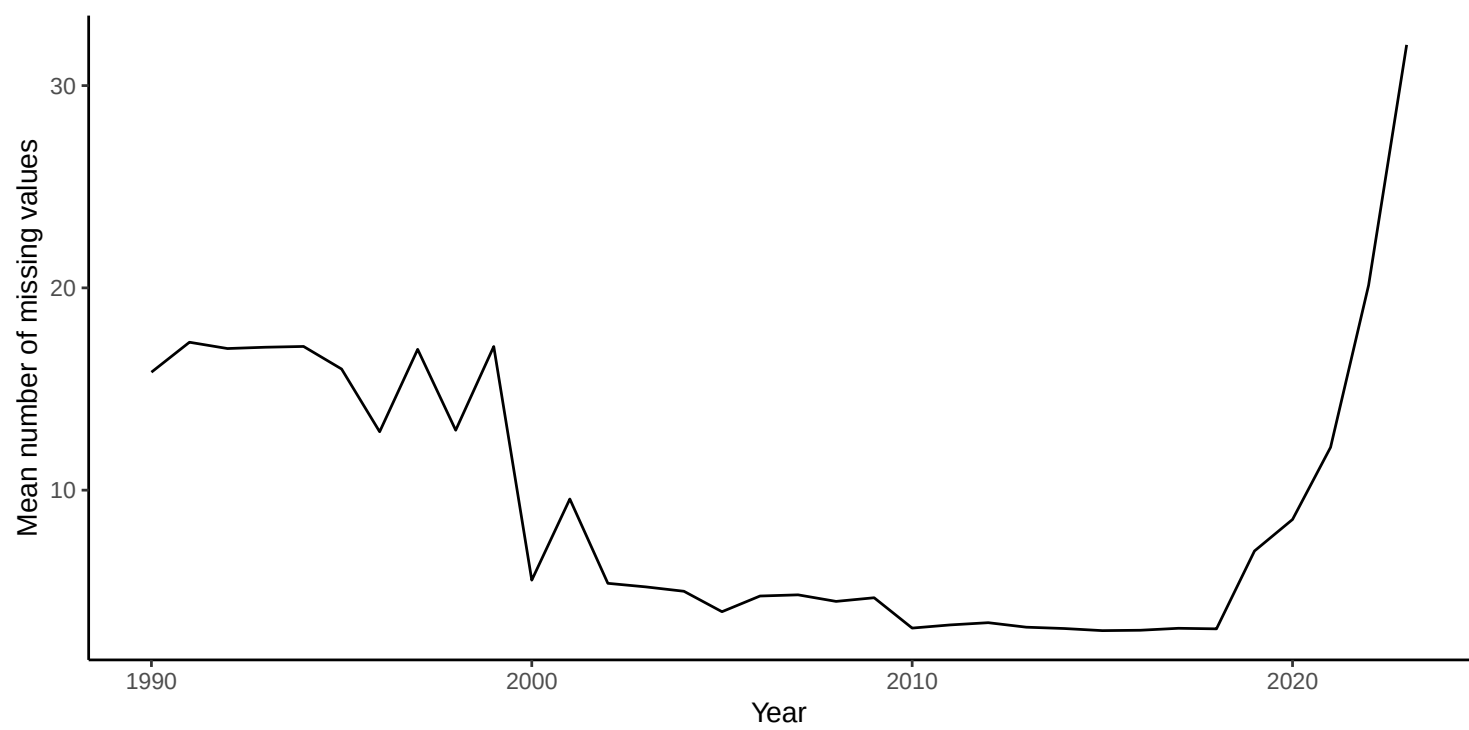


Figure S2. Percentage of predictors missing values predictors over time

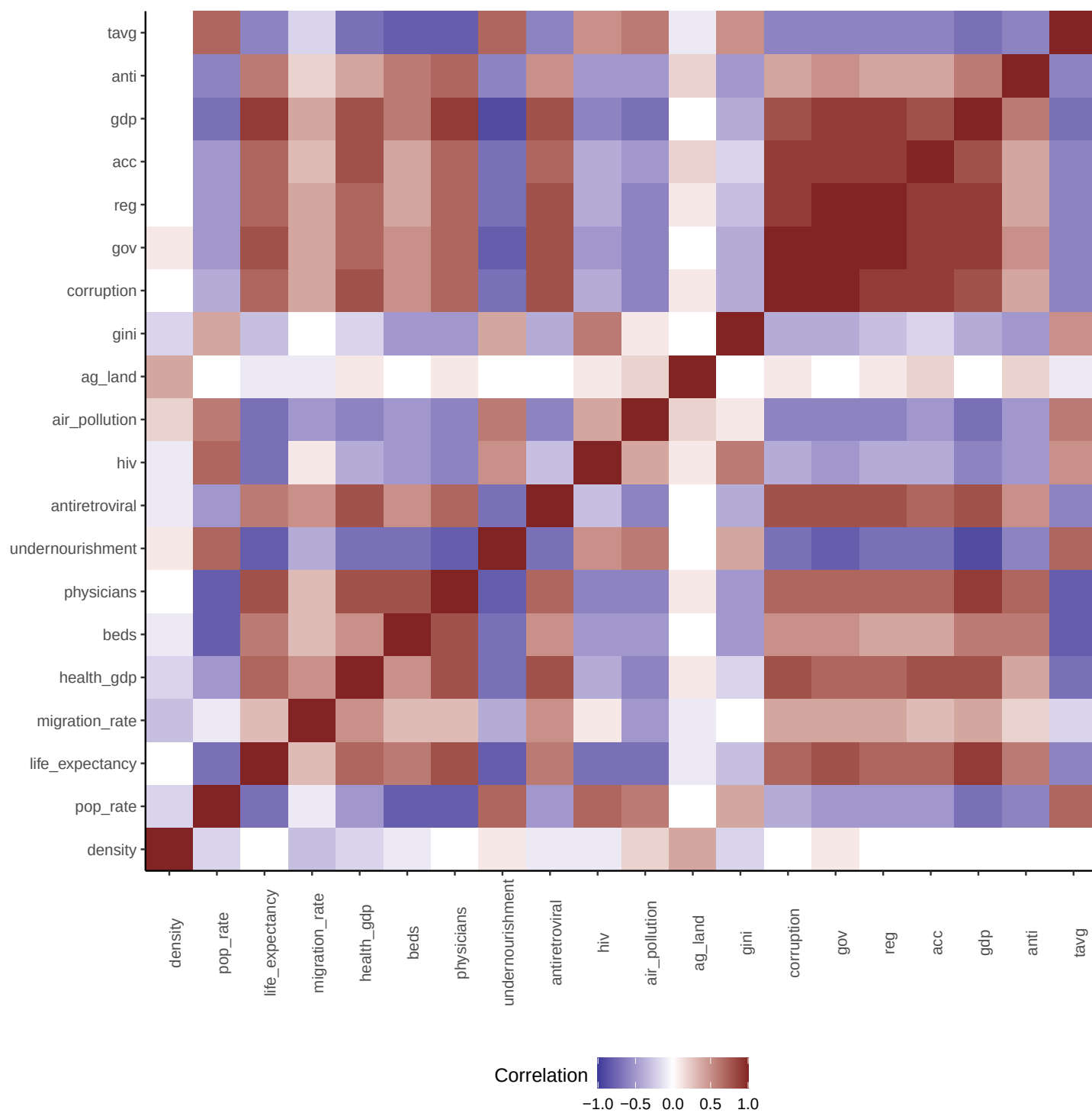


Figure S3. Pairwise correlation in predictors

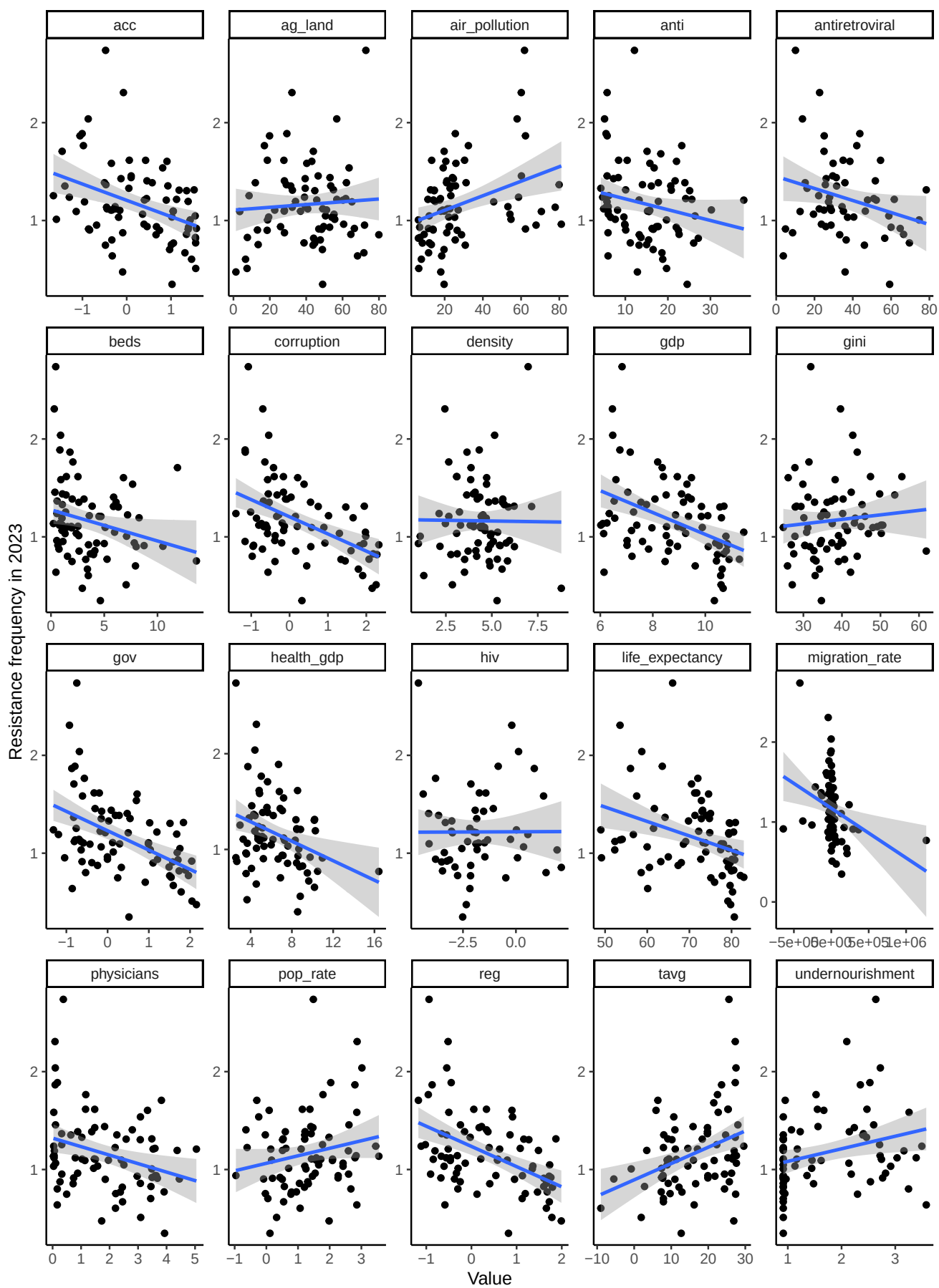


Figure S4. Univriate association between resistance and each predictor

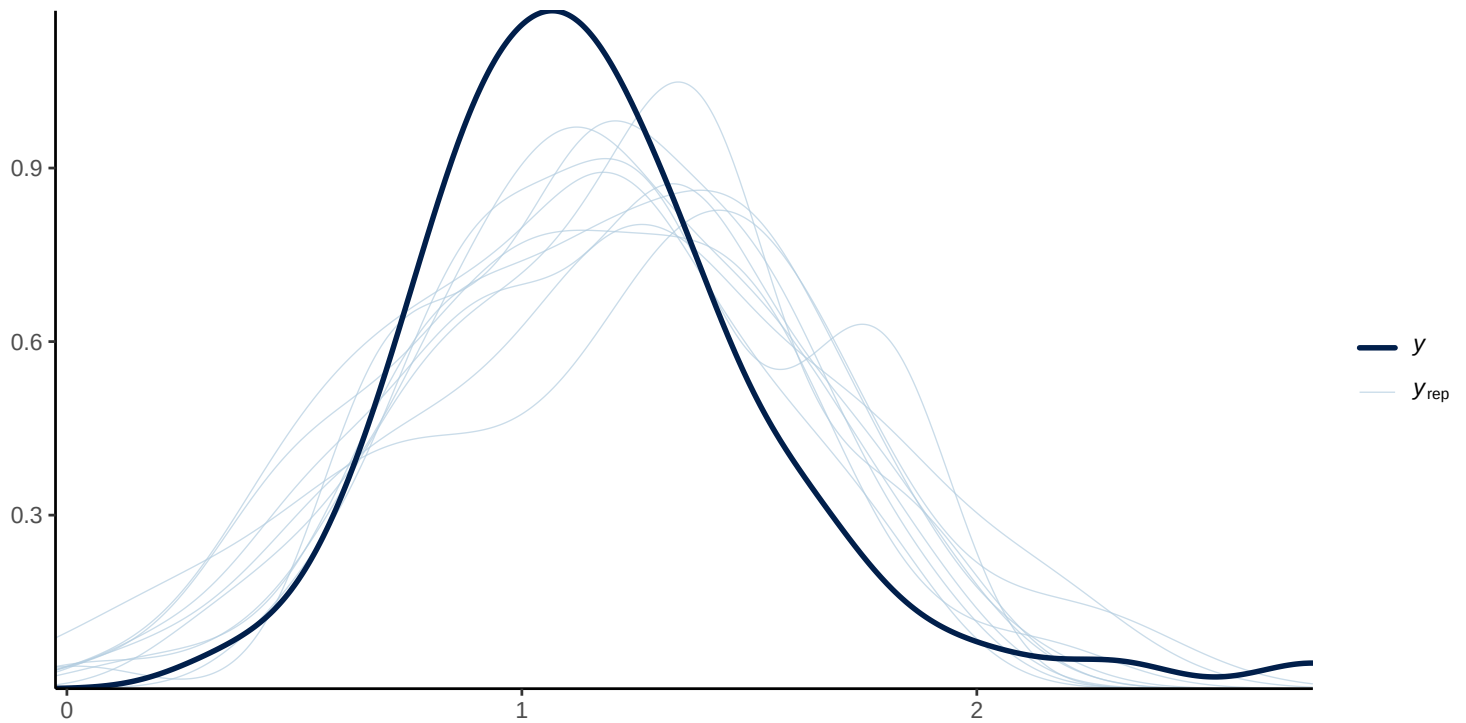


Figure S5. Posterior predictive check

NULL

Figure S6. Assumption checking

```
library(arrow)
library(brms)
library(dplyr)
library(ggrepel)

df = df_save
colnames(df)[10] = "Betalactams"

struc = read.csv("struc.csv")
struc$ind = 1
struc$id = as.character(struc$id)
#Test 1 - C. jejuni
tmp = subset(df, species == "Campylobacter jejuni")
tmp = left_join(tmp, struc)

struc_tmp = subset(tmp, ind == 1)
struc_tmp$year = struc_tmp$year - 2023 #2023 - 20 etc
range(struc_tmp$year) #2023 - 20 etc
```

[1] -20 -6

```
unique(struc_tmp$ISO.3) #GBR
```

[1] "GBR"

```

tmp = subset(tmp, !is.na(year))
tmp$year = tmp$year - 2023
tmp = subset(tmp, ISO.3 == "GBR")
tmp = subset(tmp, year >= -20)
tmp = subset(tmp, year <= -6)

struc_isolate = subset(tmp, ind == 1)
struc_isolate$uni_id = 1:nrow(struc_isolate)
nstruc_isolate = subset(tmp, is.na(ind))
nstruc_isolate$uni_id = 1:nrow(nstruc_isolate)

form =
  bf(mvbind(
    Aminoglycosides,
    Betalactams,
    Polymyxins,
    Fosfomicin,
    Glycopeptides,
    Macrolides,
    Oxazolidinones,
    Phenicols,
    Quinolones,
    Rifampicin,
    Sulphonamides,
    Tetracyclines,
    Trimethoprim) ~
    year
  ) + set_rescor(FALSE)

#fit_struc = brm(form,
#               data = struc_isolate,
#               chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#               family = bernoulli)
#saveRDS(fit_struc, "fit_struc1.rds")
fit_struc = readRDS("fit_struc1.rds")

#fit_nstruc = brm(form,
#               data = nstruc_isolate,
#               chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#               family = bernoulli)
#saveRDS(fit_nstruc, "fit_nstruc1.rds")
fit_nstruc = readRDS("fit_nstruc1.rds")

nd = tmp[c(1:15),c(10:24)]
nd$year = c(-20:-6)
nd[,c(2:14)] = NA

pred1 = posterior_linpred(fit_struc, newdata = nd, transform = T, re.form = NA)
pred2 = posterior_linpred(fit_nstruc, newdata = nd, transform = T, re.form = NA)

nam = c("Aminoglycosides", "Betalactams", "Polymyxins", "Fosfomicin", "Glycopeptides", "Macrolides", "Oxazolidin")

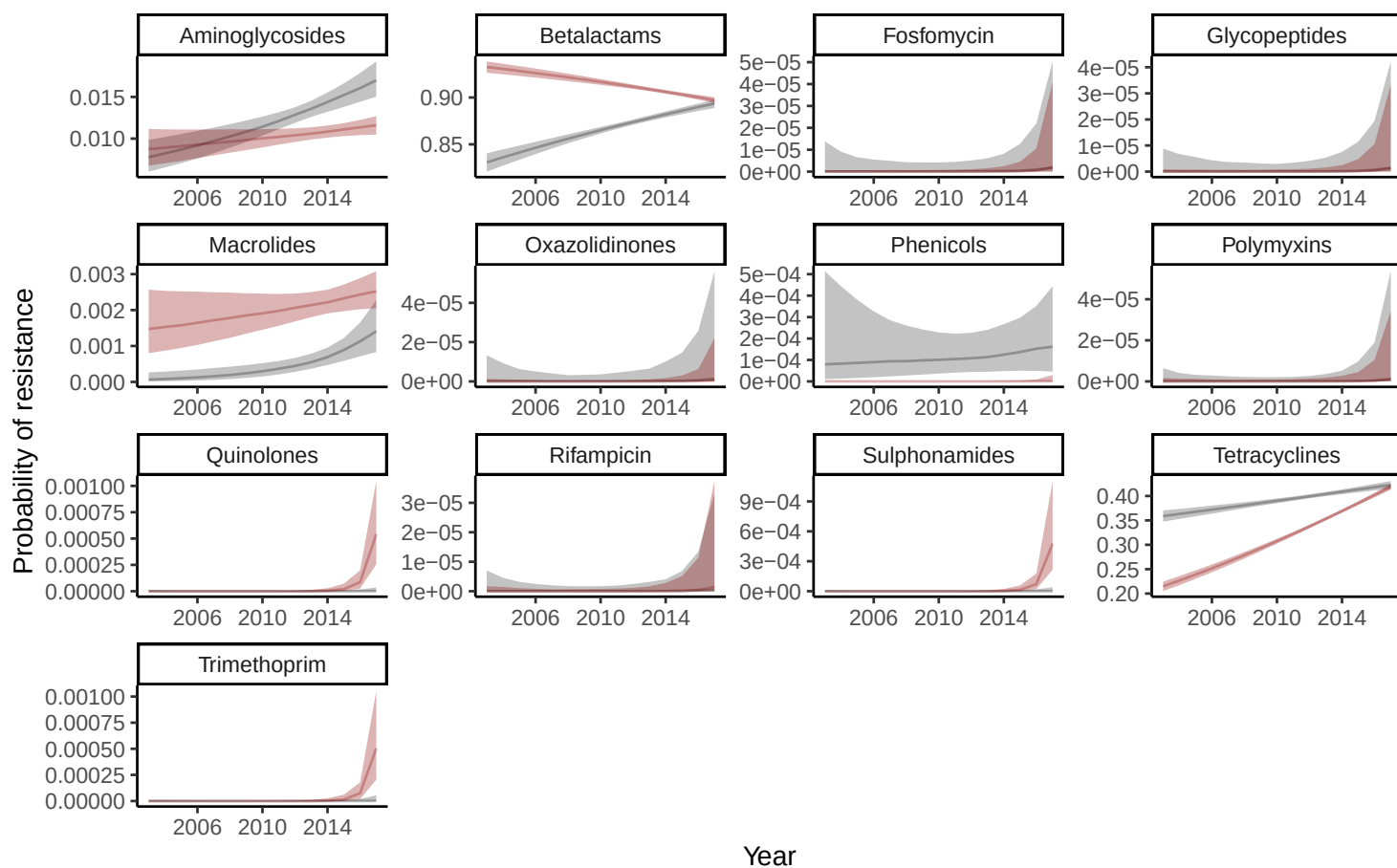
```

```

pred_cmb = NULL
pred_cmb2 = NULL
for(a in c(1:13)){
  pred1_tmp = pred1[,a]
  pred2_tmp = pred2[,a]
  sum_dat = data.frame(
    cat = nam[a],
    x = c(2003:2017),
    med1 = apply(pred1_tmp, 2, quantile, probs = 0.5),
    high1 = apply(pred1_tmp, 2, quantile, probs = 0.75),
    low1 = apply(pred1_tmp, 2, quantile, probs = 0.25),
    med2 = apply(pred2_tmp, 2, quantile, probs = 0.5),
    high2 = apply(pred2_tmp, 2, quantile, probs = 0.75),
    low2 = apply(pred2_tmp, 2, quantile, probs = 0.25))
  pred_cmb = rbind(pred_cmb, sum_dat)
}

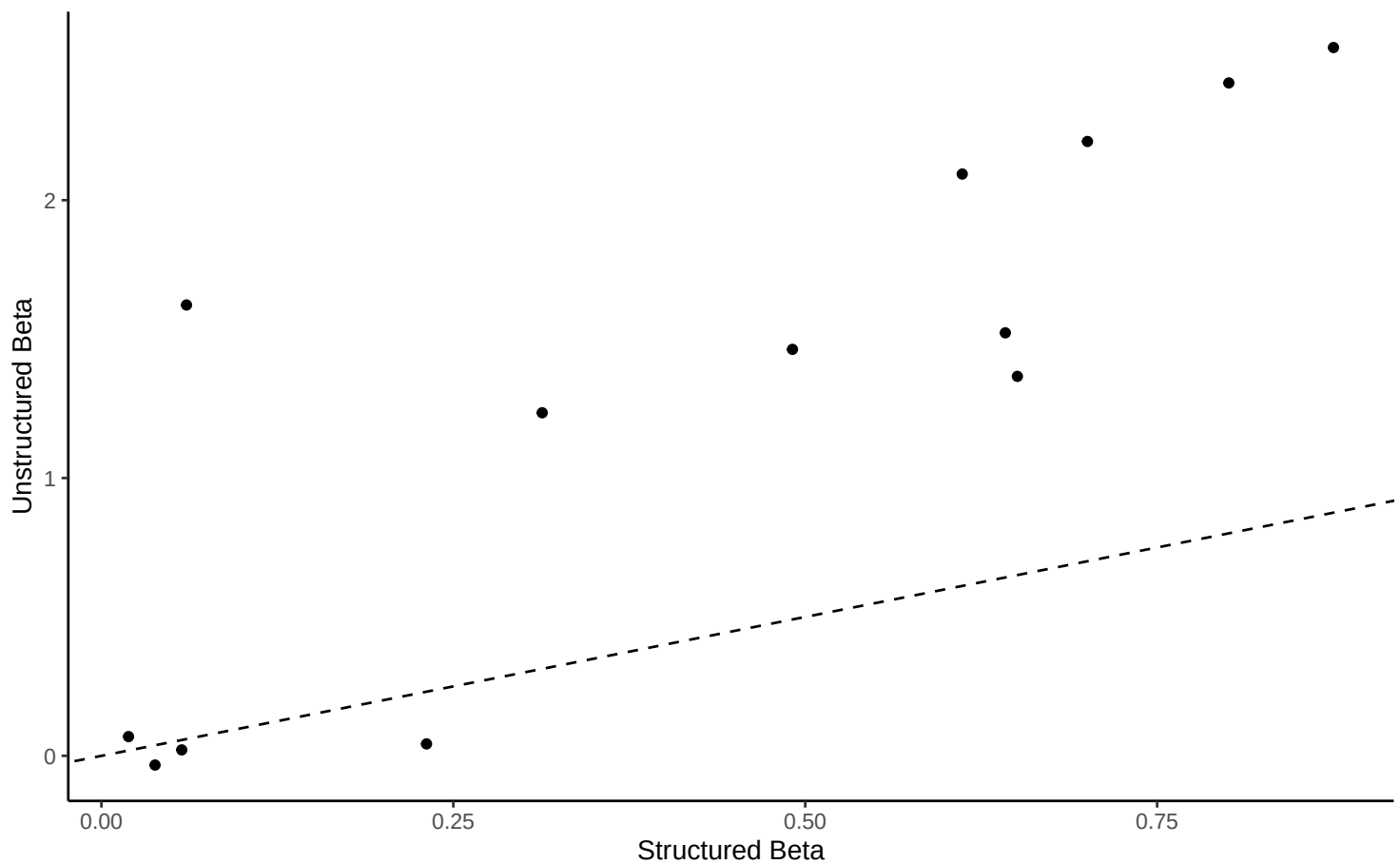
ggplot(pred_cmb) +
  geom_line(aes(x = x, y = med1), alpha = 0.3) +
  geom_line(aes(x = x, y = med2), colour = "darkred", alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low1, ymax = high1), alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low2, ymax = high2), fill = "darkred", alpha = 0.3) +
  labs(x = "Year", y = "Probability of resistance") +
  theme_classic() +
  facet_wrap(cat~., scales = "free")

```



```
slp_df = data.frame(
  fix1 = summary(fit_struc)$fixed[c(14:26),1],
  fix2 = summary(fit_nstruc)$fixed[c(14:26),1],
  names = nam)

ggplot(data = slp_df, aes(x = fix1, y = fix2)) +
  geom_point() +
  geom_abline(aes(intercept = 0, slope = 1), linetype = "dashed") +
  labs(x = "Structured Beta", y = "Unstructured Beta") +
  theme_classic()
```



```
#Test 2 - S. pneumoniae
```

```
tmp = subset(df, species == "Streptococcus pneumoniae")
```

```
tmp = left_join(tmp, struc)
```

```
tmp$ISO.3 = ifelse(tmp$country == "Morocco", "MAR", tmp$ISO.3) #Morocco country codes are incorrect and represent
```

```
struc_tmp = subset(tmp, ind == 1)
```

```
struc_tmp$year = struc_tmp$year - 2023
```

```
range(struc_tmp$year, na.rm = T)
```

```
## [1] -39 -6
```

```
unique(struc_tmp$ISO.3)
```

```
## [1] "BGD" "BLR" "BWA" "BRA" "CMR" "CAN" "CAF" "CHN" "EGY" "GMB" "GHA" "IND"
```

```
## [13] "ISR" "KWT" "MWI" "MYS" "MAR" "MOZ" "NPL" "NER" "PER" "POL" "QAT" "RUS"
```

```
## [25] "SEN" "SVN" "ZAF" "ESP" "TGO" "TTO" "USA"
```

```
tmp = subset(tmp, !is.na(year))
```

```
tmp = tmp[which(tmp$ISO.3 %in% unique(struc_tmp$ISO.3)),]
```

```
tmp$year = tmp$year - 2023 #2023 - 20 etc
```

```
tmp = subset(tmp, year >= -39)
```

```
tmp = subset(tmp, year <= -6)
```

```
struc_isolate = subset(tmp, ind == 1)
```

```

struc_isolate$uni_id = 1:nrow(struc_isolate)

#Produce a spatial covariance matrix
spa_df = unique(struc_isolate[,c("ISO.3", "centroid_lat", "centroid_lon")])
spa_df = subset(spa_df, !is.na(spa_df$centroid_lat))
spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000
spa_mat_trim = norm_range(spa_mat_trim)
spa_mat_trim = abs(spa_mat_trim - 1)
spa_id = spa_df$ISO.3
colnames(spa_mat_trim) = spa_id
rownames(spa_mat_trim) = spa_id

struc_isolate$space_id = struc_isolate$ISO.3

form =
  bf(mvbind(
    Aminoglycosides,
    Betalactams,
    Polymyxins,
    Fosfomycin,
    Glycopeptides,
    Macrolides,
    Oxazolidinones,
    Phenicols,
    Quinolones,
    Rifampicin,
    Sulphonamides,
    Tetracyclines,
    Trimethoprim) ~
    year +
    (1 + year|ISO.3) +
    (1|gr(space_id, cov = space))
  ) + set_rescor(FALSE)

#fit_struc = brm(form,
#
#           data = struc_isolate,
#           data2 = list(space = spa_mat_trim),
#           chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#           family = bernoulli)
#saveRDS(fit_struc, "fit_struc2.rds")
fit_struc = readRDS("fit_struc2.rds")

nstruc_isolate = subset(tmp, is.na(ind))
nstruc_isolate$uni_id = 1:nrow(nstruc_isolate)

#Produce a spatial covariance matrix
nstruc_isolate = subset(nstruc_isolate, nstruc_isolate$ISO.3 != "MAR")
spa_df = unique(nstruc_isolate[,c("ISO.3", "centroid_lat", "centroid_lon")])
spa_df = subset(spa_df, !is.na(spa_df$centroid_lat))
spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000
spa_mat_trim = norm_range(spa_mat_trim)
spa_mat_trim = abs(spa_mat_trim - 1)
spa_id = spa_df$ISO.3
colnames(spa_mat_trim) = spa_id

```

```

rownames(spa_mat_trim) = spa_id

nstruc_isolate$space_id = nstruc_isolate$ISO.3

#fit_nstruc = brm(form,
#                  data = nstruc_isolate,
#                  data2 = list(space = spa_mat_trim),
#                  chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#                  family = bernoulli)
#saveRDS(fit_nstruc, "fit_nstruc2.rds")
fit_nstruc = readRDS("fit_nstruc2.rds")

nd = tmp[c(1:23),c(10:24)]
nd$year = c(-28:-6)
nd[,c(2:14)] = NA

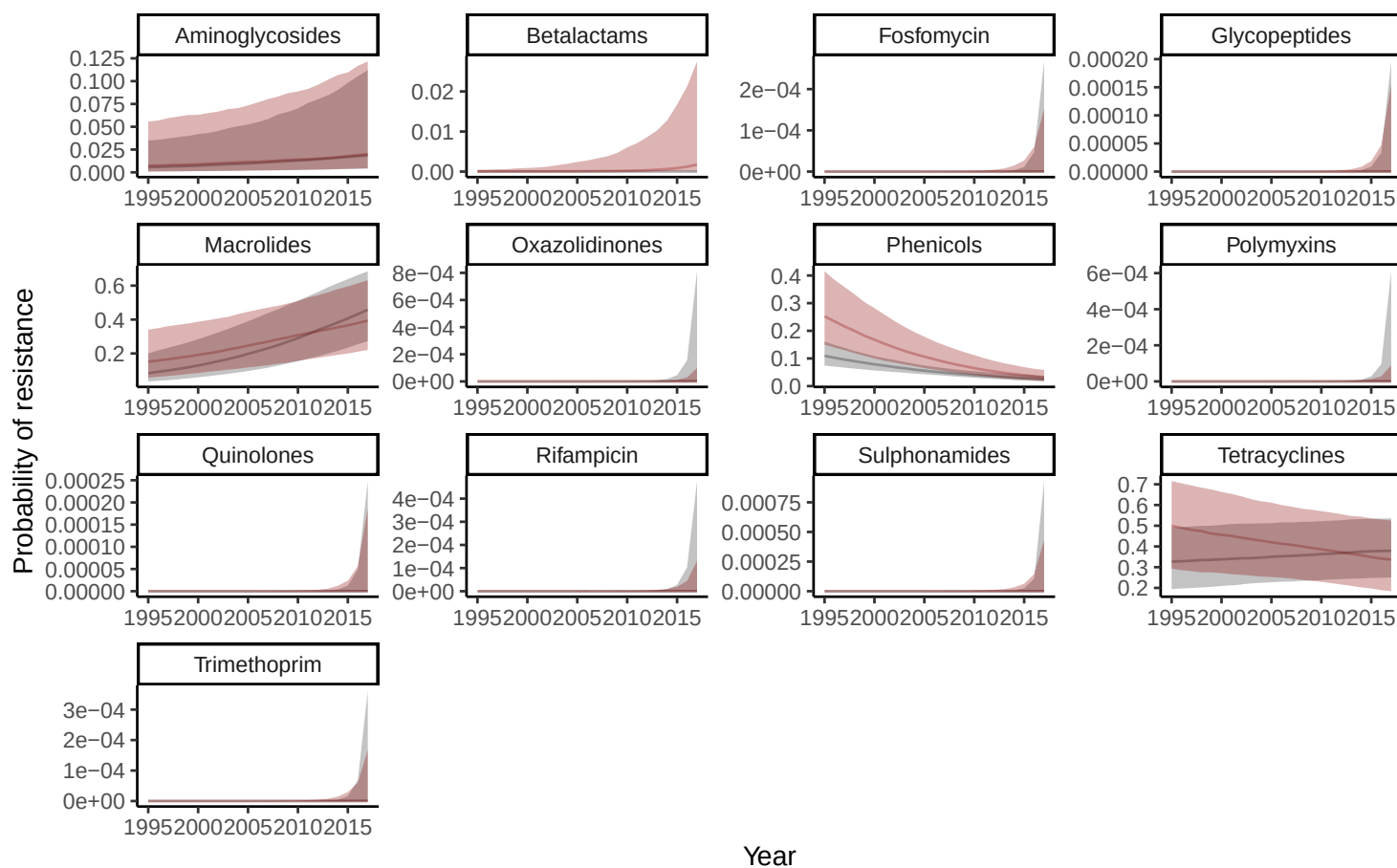
pred1 = posterior_linpred(fit_struc, newdata = nd, transform = T, re.form = NA)
pred2 = posterior_linpred(fit_nstruc, newdata = nd, transform = T, re.form = NA)

nam = c("Aminoglycosides", "Betalactams", "Polymyxins", "Fosfomycin", "Glycopeptides", "Macrolides", "Oxazolidinon")

pred_cmb = NULL
pred_cmb2 = NULL
for(a in c(1:13)){
  pred1_tmp = pred1[,a]
  pred2_tmp = pred2[,a]
  sum_dat = data.frame(
    cat = nam[a],
    x = c(1995:2017),
    med1 = apply(pred1_tmp, 2, quantile, probs = 0.5),
    high1 = apply(pred1_tmp, 2, quantile, probs = 0.75),
    low1 = apply(pred1_tmp, 2, quantile, probs = 0.25),
    med2 = apply(pred2_tmp, 2, quantile, probs = 0.5),
    high2 = apply(pred2_tmp, 2, quantile, probs = 0.75),
    low2 = apply(pred2_tmp, 2, quantile, probs = 0.25))
  pred_cmb = rbind(pred_cmb, sum_dat)
}

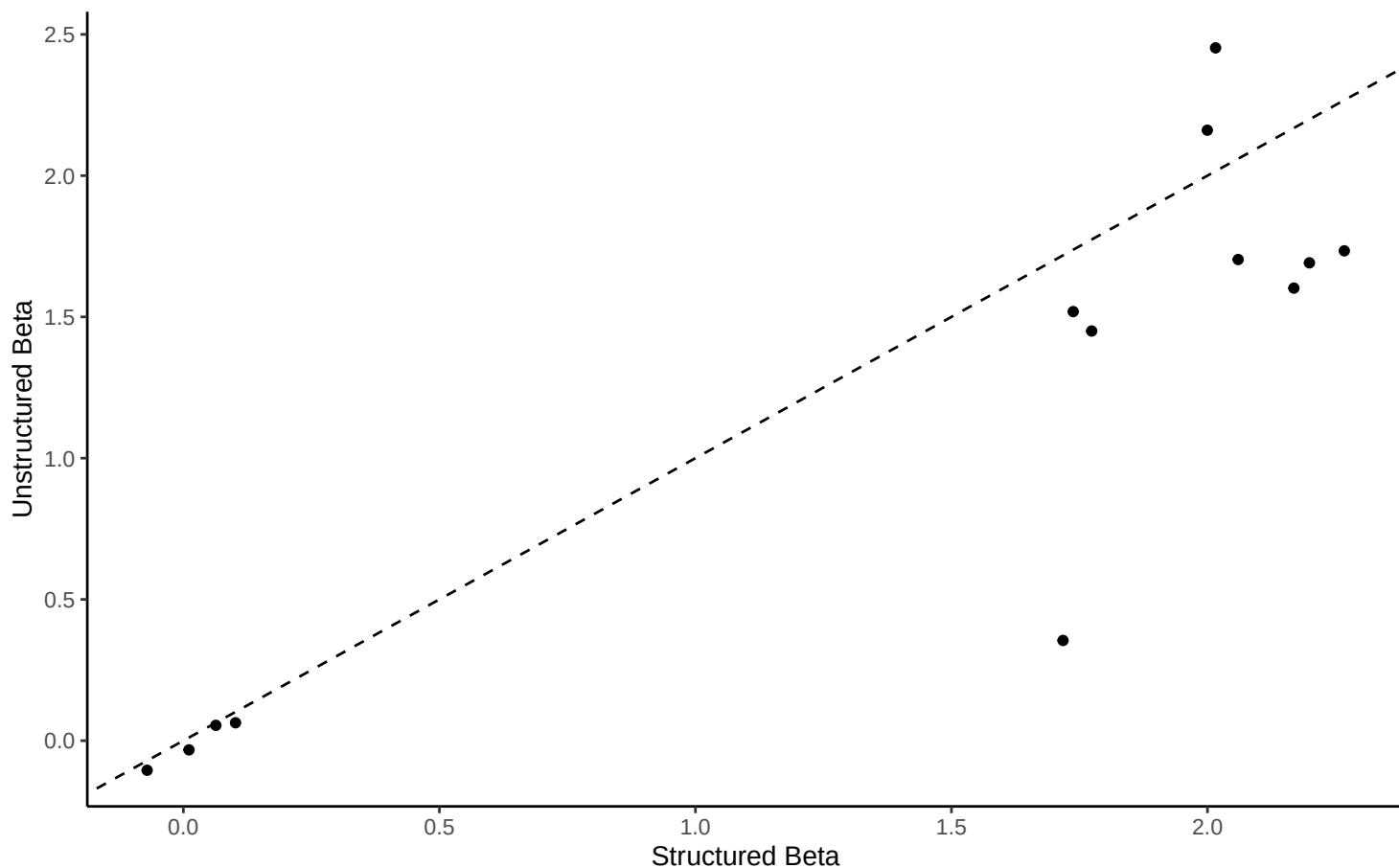
ggplot(pred_cmb) +
  geom_line(aes(x = x, y = med1), alpha = 0.3) +
  geom_line(aes(x = x, y = med2), colour = "darkred", alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low1, ymax = high1), alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low2, ymax = high2), fill = "darkred", alpha = 0.3) +
  labs(x = "Year", y = "Probability of resistance") +
  theme_classic() +
  facet_wrap(cat~., scales = "free")

```



```
slp_df = data.frame(
  fix1 = summary(fit_struc)$fixed[c(14:26),1],
  fix2 = summary(fit_nstruc)$fixed[c(14:26),1],
  names = nam)

ggplot(data = slp_df, aes(x = fix1, y = fix2)) +
  geom_point() +
  geom_abline(aes(intercept = 0, slope = 1), linetype = "dashed") +
  labs(x = "Structured Beta", y = "Unstructured Beta") +
  theme_classic()
```



```
#Test 3 - Neisseria gonorrhoeae
tmp = subset(df, species == "Neisseria gonorrhoeae")
tmp = left_join(tmp, struc)
tmp$year = tmp$year - 2023
```

```
struc_tmp = subset(tmp, ind == 1)
range(struc_tmp$year, na.rm = T) #2023 - 20 etc
```

```
## [1] -63 -5
```

```
tmp = subset(tmp, !is.na(year))
unique(struc_tmp$ISO.3)
```

```
## [1] "DZA" "ARG" "ARM" "AUS" "AUT" "BLR" "BTN" "BRA"
## [9] "BGR" "CAN" "CPV" "CHL" "CHN" "CUB" "DNK" "ECU"
## [17] "EST" "FIN" "FRA" "GMB" "DEU" "GRC" "GIN" "GNB"
## [25] "HUN" "IND" "IDN" "IRL" "ITA" "JAM" "JPN" "KEN"
## [33] "MYS" "JUL" "AGO" "SEP" "NLD" "NZL" "NOR" "PAK"
## [41] "PHL" "POL" "ROU" "RUS" "SAU" "SVK" "SVN" "ZAF"
## [49] "KOR" "ESP" "SWE" "TZA" "THA" "TUR" "UGANDA" "GBR"
## [57] "" "USA" "VNM"
```

```
struc_tmp = subset(struc_tmp, struc_tmp$ISO.3 != "UGANDA")
struc_tmp = subset(struc_tmp, struc_tmp$ISO.3 != "")
struc_tmp = subset(struc_tmp, struc_tmp$ISO.3 != "JUL")
```

```

struc_tmp = subset(struc_tmp, struc_tmp$ISO.3 != "SEP")
tmp = tmp[which(tmp$ISO.3 %in% unique(struc_tmp$ISO.3)),]
tmp = subset(tmp, year >= -33) #min 1990
tmp = subset(tmp, year <= -5)

struc_isolate = subset(tmp, ind == 1)
struc_isolate$uni_id = 1:nrow(struc_isolate)

#Produce a spatial covariance matrix
spa_df = unique(struc_isolate[,c("ISO.3", "centroid_lat", "centroid_lon")])
spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000
spa_mat_trim = norm_range(spa_mat_trim)
spa_mat_trim = abs(spa_mat_trim - 1)
spa_id = spa_df$ISO.3
colnames(spa_mat_trim) = spa_id
rownames(spa_mat_trim) = spa_id

struc_isolate$space_id = struc_isolate$ISO.3

form =
  bf(mvbind(
    Aminoglycosides,
    Betalactams,
    Polymyxins,
    Fosfomicin,
    Glycopeptides,
    Macrolides,
    Oxazolidinones,
    Phenicols,
    Quinolones,
    Rifampicin,
    Sulphonamides,
    Tetracyclines,
    Trimethoprim) ~
    year +
    (1 + year|ISO.3) +
    (1|gr(space_id, cov = space))
  ) + set_rescor(FALSE)

#fit_struc = brm(form,
#
#           data = struc_isolate,
#           data2 = list(space = spa_mat_trim),
#           chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#           family = bernoulli)
#saveRDS(fit_struc, "fit_struc3.rds")
fit_struc = readRDS("fit_struc3.rds")

nstruc_isolate = subset(tmp, is.na(ind))
nstruc_isolate$uni_id = 1:nrow(nstruc_isolate)

#Produce a spatial covariance matrix
spa_df = unique(nstruc_isolate[,c("ISO.3", "centroid_lat", "centroid_lon")])
spa_mat_trim = as.matrix(distm(spa_df[,c(3,2)], fun = distHaversine))/1000000

```

```

spa_mat_trim = norm_range(spa_mat_trim)
spa_mat_trim = abs(spa_mat_trim - 1)
spa_id = spa_df$ISO.3
colnames(spa_mat_trim) = spa_id
rownames(spa_mat_trim) = spa_id

nstruc_isolate$space_id = nstruc_isolate$ISO.3

#fit_nstruc = brm(form,
#                 data = nstruc_isolate,
#                 data2 = list(space = spa_mat_trim),
#                 chains = 4, cores = 4, iter = 1000, warmup = 500, thin = 1, refresh = 10,
#                 family = bernoulli)
#saveRDS(fit_nstruc, "fit_nstruc3.rds")
fit_nstruc = readRDS("fit_nstruc3.rds")

nd = tmp[c(1:23),c(10:24)]
nd$year = c(-28:-6)
nd[,c(2:14)] = NA

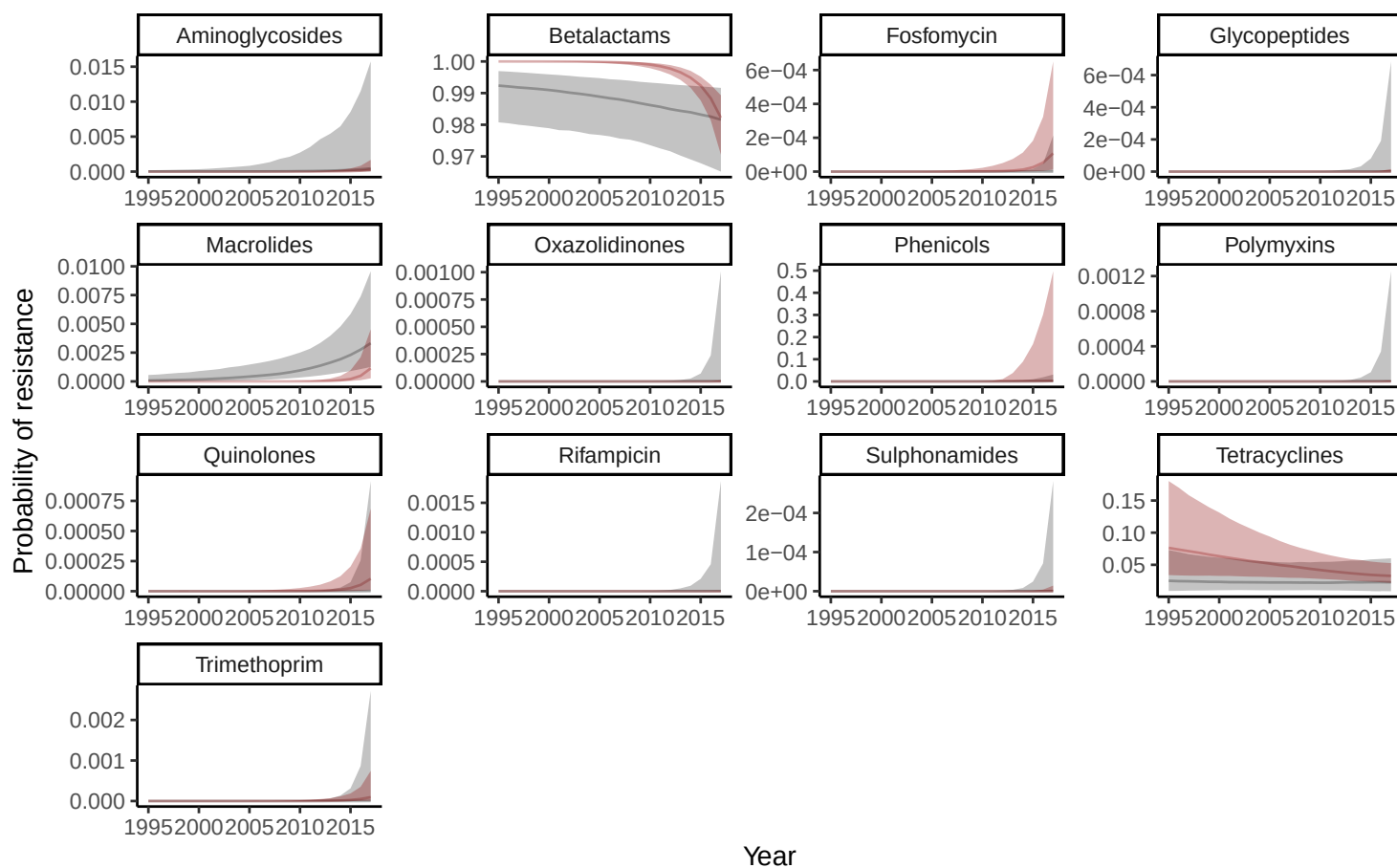
pred1 = posterior_linpred(fit_struc, newdata = nd, transform = T, re.form = NA)
pred2 = posterior_linpred(fit_nstruc, newdata = nd, transform = T, re.form = NA)

nam = c("Aminoglycosides", "Betalactams", "Polymyxins", "Fosfomycin", "Glycopeptides", "Macrolides", "Oxazolidinon")

pred_cmb = NULL
pred_cmb2 = NULL
for(a in c(1:13)){
  pred1_tmp = pred1[,a]
  pred2_tmp = pred2[,a]
  sum_dat = data.frame(
    cat = nam[a],
    x = c(1995:2017),
    med1 = apply(pred1_tmp, 2, quantile, probs = 0.5),
    high1 = apply(pred1_tmp, 2, quantile, probs = 0.75),
    low1 = apply(pred1_tmp, 2, quantile, probs = 0.25),
    med2 = apply(pred2_tmp, 2, quantile, probs = 0.5),
    high2 = apply(pred2_tmp, 2, quantile, probs = 0.75),
    low2 = apply(pred2_tmp, 2, quantile, probs = 0.25))
  pred_cmb = rbind(pred_cmb, sum_dat)
}

ggplot(pred_cmb) +
  geom_line(aes(x = x, y = med1), alpha = 0.3) +
  geom_line(aes(x = x, y = med2), colour = "darkred", alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low1, ymax = high1), alpha = 0.3) +
  geom_ribbon(aes(x = x, ymin = low2, ymax = high2), fill = "darkred", alpha = 0.3) +
  labs(x = "Year", y = "Probability of resistance") +
  theme_classic() +
  facet_wrap(cat~., scales = "free")

```



```
slp_df = data.frame(
  fix1 = summary(fit_struc)$fixed[c(14:26),1],
  fix2 = summary(fit_nstruc)$fixed[c(14:26),1],
  names = nam)

ggplot(data = slp_df, aes(x = fix1, y = fix2)) +
  geom_point() +
  geom_abline(aes(intercept = 0, slope = 1), linetype = "dashed") +
  labs(x = "Structured Beta", y = "Unstructured Beta") +
  theme_classic()
```

